HMT-TSF: A HYBRID MULTI-SCALE TEMPORAL SPATIO-FEATURE FORECASTER FOR ROBUST PUBLIC TRANSIT RIDERSHIP PREDICTION – A MALAYSIA CASE STUDY

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ABSTRACT

Predicting daily public transit ridership in Malaysia is challenged by complex spatiotemporal variables (e.g., population density, weather, and socio-economic activities) and extreme data anomalies driven by the Movement Control Order (MCO). This study identifies key ridership factors and introduces the Hybrid Multi-Scale Temporal Spatio-Feature (HMT-TSF) Forecaster. Utilizing a robust deep learning (DL) framework integrating General Transit Feed Specification (GTFS) schedules, historical weather, and Points of Interest (POIs), the proposed model is benchmarked against state-of-the-art architectures, including LSTM, BiLSTM, STGCN, and Informer. To evaluate pandemic-driven noise and predictive bias, the analysis is executed across two distinct dataset segmentations: "With MCO" (2019–2025) and "Without MCO" (2022–2025), utilizing MAE, RMSE, and R^2 evaluation metrics. Results indicate that incorporating exogenous variables significantly enhances DL predictive power over classical statistical models like ARIMA. The proposed HMT-TSF consistently outperforms baseline models across all lookback windows and dataset segmentations. Critically, training architectures on the MCO-inclusive dataset introduces severe predictive bias. Conversely, strategic data segmentation isolates these anomalies, enabling the model to accurately capture peak demands and standard urban mobility patterns. This paper provides transport authorities with an effective, data-driven methodology to optimize scheduling and close the infrastructure-to-ridership gap.

Keywords

Public Transit Reliability, Deep Learning Analytics, Ridership Forecasting, Urban Mobility Pattern Analysis

ABSTRAK

Ramalan jumlah penumpang harian pengangkutan awam di Malaysia amat mencabar disebabkan oleh pemboleh ubah ruang-masa yang rumit (seperti kepadatan penduduk, cuaca, dan aktiviti sosioekonomi) serta anomali data ekstrem akibat Perintah Kawalan Pergerakan (PKP). Kajian ini mengenal pasti faktor utama penggerak jumlah penumpang dan memperkenalkan model Hybrid Multi-Scale Temporal Spatio-Feature (HMT-TSF) Forecaster. Menggunakan rangka kerja pembelajaran mendalam (DL) teguh yang menyepadukan jadual General Transit Feed Specification (GTFS), corak cuaca sejarah, dan Tempat Menarik (POI), model yang dicadangkan ini ditanda aras berbanding seni bina terkini termasuk LSTM, BiLSTM, STGCN, dan Informer. Bagi menilai ralat (*noise*) dan bias ramalan akibat pandemik, analisis dilaksanakan merentasi dua segmentasi set data: "Dengan PKP" (2019–2025) dan "Tanpa PKP" (2022–2025), menggunakan metrik penilaian MAE, RMSE, dan R^2. Keputusan kajian menunjukkan bahawa penggabungan pemboleh ubah eksogenus meningkatkan kuasa ramalan model DL secara signifikan berbanding kaedah statistik klasik seperti ARIMA. Model HMT-TSF yang dicadangkan konsisten mengatasi prestasi model garis dasar (*baseline*) merentasi semua tetingkap imbas kembali (*lookback windows*) dan kedua-dua segmen set data. Secara kritikal, latihan model menggunakan set data inklusif-PKP memperkenalkan bias ramalan yang besar. Sebaliknya, segmentasi data yang strategik berjaya mengasingkan anomali tersebut, sekali gus membolehkan model menangkap lonjakan permintaan waktu puncak dan corak mobiliti bandar yang normal dengan tepat. Kajian ini menyediakan metodologi berasaskan data yang berkesan untuk membantu pihak berkuasa pengangkutan mengoptimumkan penjadualan serta merapatkan jurang antara peluasan infrastruktur dan jumlah penumpang.

Kata Kunci

Kebolehpercayaan Transit Awam, Analitis Pembelajaran Mendalam, Ramalan Jumlah Penumpang, Analisis Corak Mobiliti Bandar

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CHAPTER 1

INTRODUCTION

1.1 Research Background

By enhancing urban mobility where citizens prioritize public transports over private vehicles, countries can improve their economies while promoting environmental sustainability (Lu et al., 2020), aligning with United Nations Sustainable Development Goals (SDG). Malaysia, as a developing nation undergoing urbanisation, establishing an efficient public transit system is essential to achieve developed-country status (Wang et al., 2025). However, some suggested that citizens revert to private vehicles due to unreliable public transit predictions in public transit systems caused by uncertainties, weather conditions, socio-economic activities, and geopolitical events, thereby accelerates traffic congestion and air/sound pollution (Amir et al., 2025). Reliable public transit systems directly address the environmental and social challenges, enhancing residents' quality of life and mobility across metropolitan and rural areas (Wang et al., 2025).

Despite Malaysian government investments in major transit infrastructure such as mass rapid transit (MRT) Putrajaya, Shah Alam, and planned MRT Circle Line, ridership remains below average due to the uncertainties and service quality flaws (Amir et al., 2025). Prolonged travel times, untimely delays, and limited accessibility are primary causes of passenger dissatisfaction (Mee et al., 2022). These service deficiencies increase spatiotemporal complexity which conventional statistical models ARIMA fails to predict accurately. This emphasizes the important need for advanced predictive model to forecast public transit ridership accurately, consistently, and reliably.

Advancements in big data analytics and deep learning (DL) methodologies offer promising solutions in modelling intricate spatiotemporal relationships within public transit networks (Chen et al., 2022). Unlike traditional models like ARIMA, DL approaches such as LSTM and GCN integrate multiple data sources, General Transit Feed Specification (GTFS) data, historical weather patterns, Points of Interest (POIs), to fully capture multidimensional relationships (Ge et al., 2021; Chen et al., 2022). Implementing state-of-the-art (SOTA) DL architectures addresses Malaysia's current reliance on conventional models, enabling accurate ridership predictions for complex spatiotemporal data.

1.2 Problem Statement

One of the major issues with Malaysia's transit management system is predicting daily ridership accurately due to the complex relationships existing between spatiotemporal factors affecting public transport services, including population density, weather, station/stop accessibility and socio-economic activities (Amir et al., 2025). Consequently, transit authorities face critical bottlenecks in terms of scheduling, routing, and service intervals during peak hours and holidays, undermining system reliability and passenger satisfaction. Addressing this requires a dual approach: (1) identify spatiotemporal factors influencing Malaysia's ridership trends; (2) develop a robust predictive model capable of forecasting daily ridership based on these identified features.

Traditional statistical models struggle with highly fluctuating data compared to DL approaches. A critical barrier exists in historical ridership data: extreme anomalies from Malaysia's Movement Control Order (MCO) enforcement (18^{th} March 2020 to 31^{st} December 2021) produced an 85-90% passenger drop, creating highly noisy patterns unrepresentative of normal operations (Amir et al., 2025). Training AI models on such extreme outlier data introduces significant predictive bias, as pandemic-driven behaviour differs fundamentally from typical urban mobility patterns (Zhao & Lan, 2025). Models trained on contaminated datasets often struggle to recognize true ridership peaks once daily life normalizes (Lv et al., 2021). Therefore, this researcher strategically compare two datasets – one including and one excluding the MCO period – to determine whether this anomaly should be treated as noise before model training.

1.3 Research Questions

1) What are the primary spatiotemporal factors (e.g. weather, POIs, population density) that heavily affect public transit ridership trends in Malaysia?
2) How accurately and consistently can existing DL architectures (spatiotemporal-based, graph-based, and attention-based) compare against each other and proposed model in predicting Malaysia's public transit ridership?
3) How does the inclusion of extreme data anomalies, specifically the Malaysia's MCO period, impact the predictive accuracy and bias of existing DL models and proposed model?

1.4 Research Objectives

- To identify and categorize the primary spatiotemporal factors affecting Malaysian public transit ridership through a systematic analysis of historical performance data.
- To develop Hybrid Multi-Scale Temporal Spatio-Feature (HMT-TSF) Forecaster that integrates Malaysia's spatiotemporal data to estimate future ridership trends.
- To benchmark the proposed model against existing baseline, spatiotemporal-based, graph-based and attention-based models using combined score and R^2.

- To evaluate and compare model performance across two distinct dataset segmentations (with and without MCO period) to determine the impact of extreme outlier data on predictive bias.

1.5 Research Scope

Table 1.1

Methodology, Metrics, Models, and Malaysia's MCO Period Used in This Research

Aspect	Description	
Models*	Baseline	LSTM
	Spatiotemporal	BiLSTM, ST-LSTM, CNN-LSTM, CNN-BiLSTM, TPA-LSTM
	Graph	STGCN, ASTGCN, PDR-STGCN, STSGCN, STFGNN, MTGNN
	Attention	Autoformer, Informer
	Proposed	Hybrid Multi-scale Temporal Spatio Feature (HMT-TSF) Forecaster
Evaluation Metrics	Combined MAE, RMSE and MAPE, and R^2	
Geographical Focus	Bus, urban rail, commuter transit networks with available data throughout whole Malaysia.	
MCO Period Segmentation	Without MCO	1st Jan 2022 to 31st Dec 2025
	With MCO	1st Jan 2019 to 31st Dec 2025

Note. *. Models full form refer to Appendix C.

1.6 Research Contribution

- Provide a detailed comparative analysis evaluating multiple deep learning models across distinct lookback windows, explicitly segmented by MCO periods.
- Introduce a predictive model that outperforms baseline SOTAs across all lookback windows and MCO inclusion/exclusion, accompanied by a reduced-feature variant based on SHAP-based feature importance ranking, to evaluate the removal of near-zero important spatiotemporal features toward ridership prediction.
- Develop a command-line interface (CLI) inference to predict 7-day ridership for 12 distinct transit networks based on historical distribution percentages with the proposed model.

1.7 Thesis Organization

Chapter 2: Literature Review

Reviews public transport reliability, ridership dynamics, and the Malaysian transit context. It covers transit data ecosystems, preprocessing challenges, and the shift from traditional models to deep learning for spatiotemporal forecasting and closes by identifying the research gap.

Chapter 3: Methodology

Describes the full pipeline: eight data sources, feature engineering, EDA, and sequence construction under MCO-inclusive and MCO-exclusive settings. It presents fourteen baseline models and the proposed HMT-TSF architecture, together with the Combined% evaluation framework.

Chapter 4: Findings & Results

Reports the comparative evaluation of fifteen architectures across default and tuned configurations, three look-back windows, and both MCO conditions. It covers baseline rankings, tuning and look-back sensitivity, MCO robustness, fit diagnostics, HMT-TSF performance, SHAP analysis, and the feature-reduced HMT-TSF-FR variant.

Chapter 5: Discussion & Conclusion

Interprets the findings, discusses implications for Malaysian transit planning, acknowledges study limitations, outlines future work, and concludes on the contribution of HMT-TSF to demand-responsive operations.

CHAPTER 2

LITERATURE REVIEW

2.1 Public Transport Reliability and Demand Dynamics

2.1.1 Defining Reliability: Metrics and User Perception

Public transport reliability is a multi-dimensional construct that integrates objective operational performance indicators with subjective passenger experience. A study defines travel time reliability as the calculated time gap between expected and actual travel times, citing route length, signalized intersections, and departure delays as primary structural causes of network unreliability (Mohamed et al., 2021). Conversely, some researchers also demonstrate that user satisfaction within the Klang Valley Light Rail Transit (LRT) system is mediated by “soft” reliability factors, the provision of accurate real-time information and the quality of station amenities (Ibrahim et al., 2022).

This divergence indicates that while transit agencies like RapidKL traditionally prioritize “hard” technical metrics such as headway adherence and on-time performance, the passenger’s psychological perception of reliability is shaped by the immediate physical environment and the mitigating effect of transparent communication. For instance, structural breakdowns and delays on the LRT Kelana Jaya line, alongside extreme disruptions like the derailment near Chan Sow Lin station on LRT Ampang-Sri Petaling line, highlight that operational unpredictable execution converts directly into acute user stress and anxiety.

Consequently, understanding the intersection of technical metrics and subjective factors is vital for establishing standardized evaluation criteria, as mapped in the comprehensive breakdown of operational, informational, and service quality parameters detailed in Table 2.1.

Table 2.1

Comparison of Technical Reliability Metrics and User Perception Factors

Metric Category	Technical[a]	Subjective[b]
Temporal (Mohamed et al., 2021)	Headway adherence, Travel time variability	Punctuality, Wait time anxiety
Informational (Rezapour & Ferraro, 2021)	PIS accuracy, Real-time data latency	Certainty, Reduced stress
Service Quality (Ibrahim et al., 2022)	Vehicle cleanliness, Amenity availability	Comfort, Perceived safety
Operational (Sogbe et al., 2024)	Frequency, Route coverage	Accessibility, Convenience

Note. a. Operational Indicator. b. User Perception

2.1.2 The Relationship between Ridership and Service Unreliability

Chronic service unreliability generates a sever demand-erosion cycle that fundamentally undermines public transit ridership. The absence of reliable predictions regarding traffic and weather variations causes citizens to abandon public transit in favour of private vehicles (Amir et al., 2025), supported with another finding which identify prolonged travel times and untimely delays as the leading causes of passenger dissatisfaction in metropolitan areas like Kuala Lumpur and Manila (Mee et al., 2022).

When operational variance grows, passengers are forced to allocate substantial “buffer time” into their personal schedules to insure against systemic delays. For most commuters, this continuous temporal penalty quickly outweighs the financial savings of public transport, causing stagnant or declining model shares even during periods of rising fuel costs or high private vehicle ownership penalties, as observed in decentralized urban sectors such as Kota Samarahan. This behavioural shift establishes a self-reinforcing loop where transit unreliability drives private vehicular growth, thereby intensifying the road congestion that further degrades transit schedule adherence, as visually conceptualized in the behavioural cycle shown in Figure 2.1.

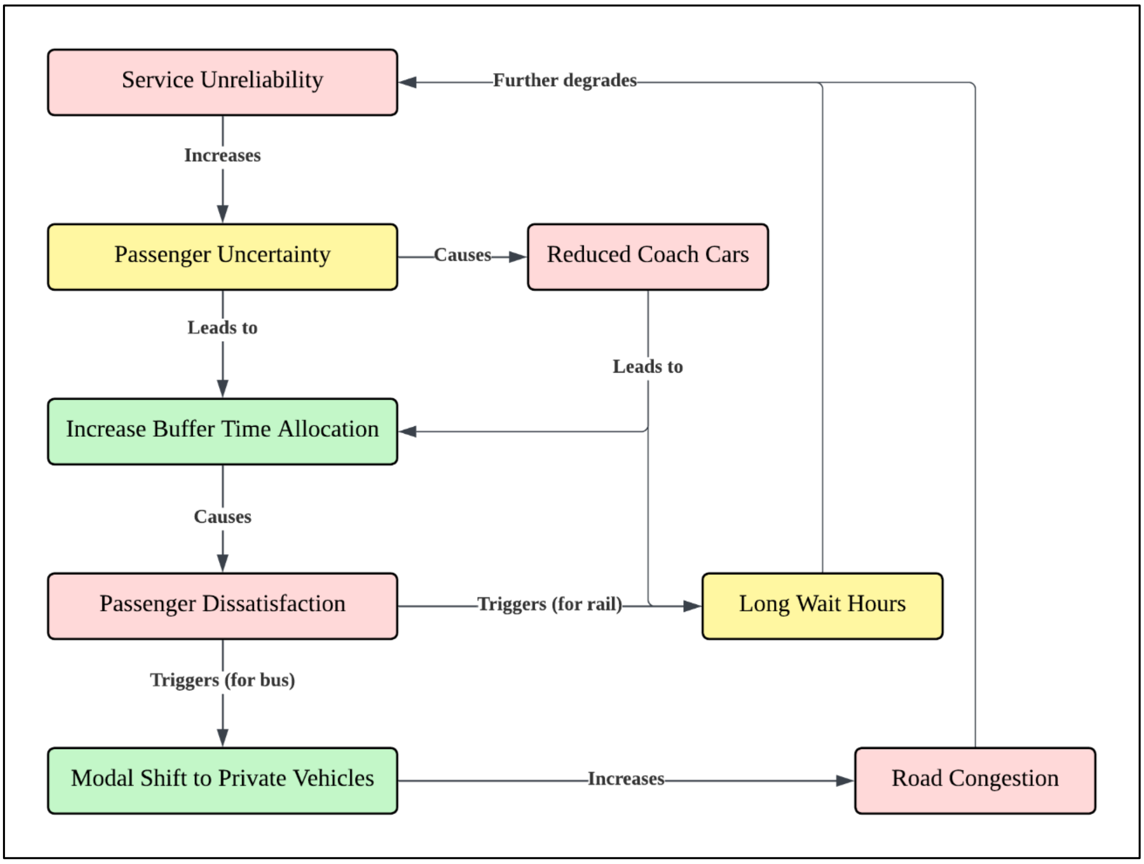

Figure 2.1 - Vicious Cycle of Transit Unreliability and Ridership Uncertainty

2.1.3 The Landscape of Public Transit in Malaysia

The Malaysian public transportation sector is currently defined by a sharp mismatch between extensive capital infrastructure investments and persistently low commuter modal shares. Despite massive government-funded transit expansions – including the Mass Rapid Transit (MRT) Putrajaya line, Shah Alam line, and the upcoming MRT3 Circle line – national public transport usage remains stagnant at approximately 20%, falling significantly short of the 40% target established by the National Transport Policy

(NTP) 2019 – 2030. Research highlights that in regions like Sarawak, private vehicle ownership remains exceptionally high due to the perceived inadequacy of bus service quality regarding tangibility and operational responsiveness (Ubaidillah et al., 2022).

This disparity proves that physical network expansion is sufficient on its own to shift commuter habits; infrastructure must be supported by localized reliability and data-driven operational optimizations to resolve localized service gaps. While programs like the MyBas initiative in Johor Bahru and Penang have successfully leveraged GTFS data to optimize networks and capture over 14.5 million trips, systemic barriers like the "first-mile-last-mile" (FMLM) connectivity gap continue to limit the efficacy of the broader national network. Resolving these localized structural inefficiencies requires transit management to pivot away from purely physical expansion toward DL methodologies capable of accurately forecasting spatiotemporal ridership to mitigate daily unreliability.

2.2 Transit Data Ecosystems and Anomalies

2.2.1 Sources of Transit Data

Modern transit data ecosystems combine diverse endogenous operational streams and exogenous environmental inputs to form a comprehensive foundation for urban mobility analysis. Two studies founded to classify core endogenous sources as Automated Fare Collection (AFC) systems for ridership tracking, Automated Vehicle Location (AVL) systems for real-time GPS tracking, Automated Passenger Counting (APC) sensors, and static GTFS schedules (Ge et al., 2021; Liu et al., 2025). Concurrently, another study also defines critical exogenous variables as real-time weather metrics, road network traffic data, POIs extracted via land-use profiles, and regional socio-economic demographics (Amir et al., 2025).

Synthesizing these multi-source streams into an integrated matrix, as categorized by the operational features, collection frequencies, and specific analytical applications outlined in Table 2.2 and 2.3 for endogenous and exogenous data, respectively, is a vital prerequisite for advanced predictive modelling.

Table 2.2

Endogenous Data Sources: Examples, Applications, and Characteristics

Specific Examples	Typical Applications	Key Characteristics
AFC Systems	Ridership analysis, origin-destination, revenue tracking	Transactional, passenger-centric
AVL Systems	Real-time tracking, schedule adherence, fleet management	Spatiotemporal, vehicle-centric
APC Systems	Occupancy rates, detailed ridership statistics	Discrete counts, vehicle-centric
Vehicle Sensors	Maintenance, operational efficiency	Real-time, diagnostic
GTFS Data	Route planning, schedule information, app development	Static, foundational

Table 2.3

Exogenous Data Sources: Examples, Applications, and Characteristics

Specific Examples	Typical Applications	Key Characteristics
Weather Data	Demand forecasting, operational adjustments	Environmental, dynamic

Traffic Data	Travel time prediction, congestion management	Real-time, network-wide
POIs	Demand modelling, urban planning	Static/dynamic, location-based
Demographic/Socio-economic	Long-term planning, demand forecasting	Static/slow-changing, population-centric

2.2.2 Data Quality, Noise and Preprocessing Challenges

Raw transit data ecosystems are highly vulnerable to fragmented quality issues and severe noise, demanding extensive preprocessing before deployment in DL models. One research points out that public transport agencies frequently collect disjointed and fragmented data streams that impede comprehensive system interpretation (Keller et al., 2022), while another identifies pervasive data quality threats such as sensor omissions, localized GPS telemetry errors, formatting inconsistencies across legacy systems, and extreme outliers (S.K.B et al., 2024).

These imperfections introduce major challenges during data cleaning, multimodal integration, feature transformation, and data reduction stages. For example, a missing data gap caused by an IoT network outage could generate significant downstream predictive bias if it left unaddressed, while the lack of alignment across spatial resolutions creates high noise that obscures actual demand signals. Because raw transit data is rarely usable in its initial state, implementing rigorous, automated preprocessing and feature engineering pipelines is essential to ensure data integrity and unleash the full potential of advanced analytics.

2.2.3 Extreme Anomalies in Spatiotemporal Data

Spatiotemporal transit profiles are sensitive to extreme, non-typical anomalies that degrade the reliability and generalization capabilities of predictive models. Extreme anomalies deviate drastically from standard mobility baselines, exemplified by Malaysia's MCO during the COVID-19 pandemic, which caused an abrupt 85-90% ridership drop across all transit networks (Maria-Arribas et al., 2026). Training predictive models directly on these contaminated datasets introduces severe predictive bias because pandemic-era containment behaviours differ fundamentally from normal urban mobility patterns (Zhao & Lan, 2025).

When neural network architectures ingest uncorrected outlier sequences, they risk overfitting to anomalous troughs, confusing regular seasonal patterns, and failing to recognize normal ridership peaks once standard daily operations resume. Traditional anomaly detection methods relying on basic statistical thresholds often fail when applied to high-dimensional spatiotemporal data, driving the need for advanced techniques like ensemble DL or strategic dataset segmentation. This structural vulnerability underlines the necessity of this study's approach: splitting the time-series data into distinct MCO-inclusive and MCO-exclusive segments to empirically evaluate and isolate the impact of extreme noise on DL architectures.

2.3 Evolution from Traditional to DL Models

2.3.1 Traditional Architectures

Traditional statistical and machine learning architectures offer strong model interpretability but are fundamentally limited by their inability to capture the non-linear, multi-dimensional complexities of modern transit data. Historically, forecasting models have relied on classical frameworks like ARIMA, Historical Average (HA), Vector

AutoRegressive (VAR) models, Support Vector Regression (SVR), and Random Forest Regression (RFR). These traditional statistical architectures demand strict data stationary and rigid distribution assumptions, which directly conflict with highly fluctuating, time-varying traffic data (Yin et al., 2022).

The structural limitations detailed in the comparative breakdown of data handling, assumptions, and scalability metrics in Table 4 emphasize the clear need to transition away from rigid classical models toward highly flexible DL frameworks (Sobrie et al., 2023).

Table 2.4

Comparison of Traditional and DL Models

Aspect	Traditional Models	DL Models
Data Handling	Small-scale, structured	Large-scale, multi-source
Dependencies	Linear, short-term	Non-linear, long-term
Complexity	Low (functional forms)	High (hierarchical features)
Assumptions	Strict (Stationarity, normality)	Minimal (Data-driven)
Interpretability	High	Lower (Black-box nature)
Scalability	Limited	High

2.3.2 Deep Learning Approaches in Spatiotemporal Transit Data

The main advantage of using DL architectures is that they can circumvent many restrictions posed by the conventional models by means of automatically capturing hierarchies of non-linear spatiotemporal features from transit multi-source data. The most efficient modern DL frameworks incorporate specially designed neural building blocks, Convolutional Neural Networks (CNNs) capture spatial correlations from grid structures, Recurrent Neural Networks (RNNs) like LSTMs track sequential temporal dependencies, Graph Convolutional Networks (GCNs) model non-Euclidean network topologies, and attention mechanisms dynamically weigh feature significance. For instance, multi-model GeoSpatial-Temporal LSTM (GT-LSTM) that achieved a 15% reduction in MAPE and a 20% reduction in RMSE over traditional methods and standard DL baselines (S.K.B et al., 2024).

These architectures eliminate the necessity for explicit feature engineering through the learning of multi-dimensional interactions directly from the data. The use of transportation networks as connected graph nodes and environmental variables as dynamic weights, models like STGCN, ASTGCN, and TPA-LSTM can simultaneously process structured data (such as GTFS) and unstructured covariates (such as weather and POIs). Successful deployment of these diverse DL architectures requires establishing an equally rigorous benchmarking methodology to benchmark their respective accuracy and error.

2.3.3 Benchmarking Methodologies and Evaluation Metrics

Rigorous, standardized benchmarking methodologies and multi-metric validation frameworks are essential to verify the accuracy, consistency, and operational reliability of predictive models. In comprehensive DL surveys, some studies utilize MAE, RMSE, and MAPE as primary evaluation metrics (Jiang et al., 2021), while others emphasize the importance of pairing relative and absolute indices to validate SOTA results on benchmark datasets (Mystakidis et al., 2025).

Evaluating models through a single metric can mask critical performance flaws. A comprehensive benchmarking framework, as shown in Table 2.5, must combine comparative analysis against established baselines, sensitivity analysis under varying data densities, and robustness testing across distinct geographical and anomalous contexts. Adhering to these structured metrics ensures a fair comparison between 14 baseline architectures evaluated in this study, providing a clear path to validating the performance gains of the proposed HMT-TSF forecaster.

Table 2.5

Common Evaluation Metrics and Their Focus (Jiang et al., 2021; S.K.B et al., 2024; Mystakidis et al., 2025)

Metric	Focus
MAE	Treats all errors equally and highly robust to outliers
RMSE	Heavily penalized large errors and sensitive to outliers
MAPE	Measures error relative to actual values, easy to understand
R^2	Explanatory power, how well model predict based on given data

2.4 AI-Driven Operational Strategies and Sustainability

2.4.1 Proactive Scheduling and Resource Allocation

Transitioning from reactive transit management to AI-driven predictive modelling enables transit agencies to implement proactive scheduling and highly optimized resource allocation. Public transport networks are evolving through AI's capacity for data-driven optimization (Jevinger et al., 2024). A study proved that LSTM encoder-decoder architecture outperforms traditional rule-based business systems by 18% to 23% in real-time delay forecasting (Sobrie et al., 2023). Furthermore, reports also show that AI-powered planning platforms maximize bus schedule efficiency and fleet deployment by anticipating changing user demands (Levner, 2025; CCS Global Tech, 2026).

Traditional dispatch scheduling is reactionary in nature; changes in service frequencies are made only after delays or disruptions have occurred in the system due to over-crowding or other reasons. On the contrary, DL models can predict passenger surges and bottleneck formations well in advance, which allows for dynamic changes in service intervals as well as coach allocation based on future passenger demand. Their successful deployment depends on overcoming several technical and organizational barriers within transit agencies.

2.4.2 Deployment Challenges of AI in Transit Management

AI solution adoption into public transportation system framework is hampered by multiple obstacles including data quality, infrastructure scale, and organizational resistance. Fragmented corporate data structures and internal organizational silos frequently block the adoption of advanced data science methods (Keller et al., 2022), whereas two evidence shows that high upfront implementation costs, strict data privacy compliance, and stakeholder resistance to operational changes as major adoption barriers (Suwaidi et al., 2022; Hashimzai & Mohammadi, 2024).

DL approaches need large-scale, clean datasets, meaning there must be considerable long-term commitment in data governance and scalable computing infrastructure. Without clear data integration strategies and collaborative pipelines connecting software developers with transit operators, such complex predictive systems will just be theoretical endeavours. It is crucial to overcome these multifaceted organizational obstacles since addressing them will allow transit agencies to unlock the significant environmental benefits associated with such AI-based optimizations, as depicted in the cause-and-effect flow in Figure 2.2.

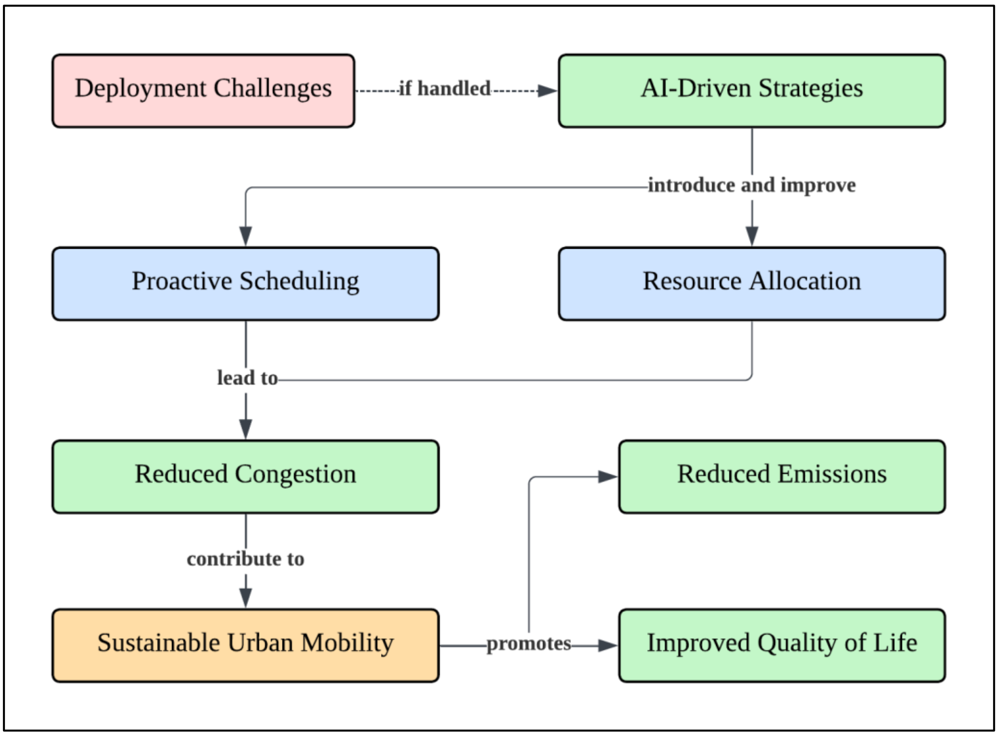

Figure 2.2 - Cause-and-effect of Resolving AI Deployment Challenges in Public Transit Systems

2.4.3 Sustainable Urban Mobility and Environmental Impact

Implementing AI optimizations within public transit networks directly supports sustainable urban mobility gaols by reducing operational waste and mitigating environmental degradation. Transforming urban mobility is vital for lowering global transportation emissions (Son et al., 2025) whereas AI-driven regularizations of headways and schedule adherence minimize fuel waste, emergency maintenance costs, and operational inefficiencies (Vujadinovic et al., 2024).

If predictive models prove successful in improving transit efficiency and reducing waiting time, they will help expedite the modal shift that involves choosing public transits over private vehicles. The combination of this modal shift along with the improvement of bus idling and rail routes directly lowers urban emission rates and improves overall air quality (CCS Global Tech, 2026). The optimization of transit operations through AI-powered forecasts serves as a vital tool for advancing comprehensive urban sustainability and improving the long-term quality of life for citizens.

2.5 Research Gap

Existing literature lacks a hybrid ridership forecaster that integrates multi-scale spatiotemporal features while explicitly isolating predictive bias from severe structural breaks such as Malaysia's MCO period. Although recent studies apply deep learning to public transport ridership (Farahmand et al., 2023; Amir et al., 2025), they rely on standard single-paradigm architectures that do not jointly model multi-scale temporal dynamics and multi-dimensional spatial interactions. Pandemic-era anomalies further confound model training (Maria-Arribas et al., 2026), yet no Malaysian study systematically quantifies how such breaks distort graph-based and attention-based state-of-the-art models. Excluding the MCO period disrupts series continuity; retaining it uncorrected introduces predictive bias, and without a unified benchmark across sequential, graph, and attention paradigms under both MCO conditions, operators lack evidence for selecting robust forecasters. Table 2.6 consolidates these limitations across prior work. This study addresses them through the HMT-TSF, enabling a shift from reactive to proactive, data-driven transit management.

Table 2.6

Comparative Summary of Prior Forecasting Studies Supporting the Research Gap

Study	Scope & Target	Model & Data	Limitations	Gap Link
(Farahmand et al., 2023)	Netherlands (bus network); Daily bus ridership, weather-conditioned	Standard DL (CNN/LSTM family); Weather covariates only	✗ Single temporal scale; ✗ No pandemic/MCO segmentation; ✗ Single-model study	Applies baseline DL to ridership but not multi-scale spatiotemporal fusion or anomaly isolation
(Amir et al., 2025)	Malaysia (Rapid Bus Kuantan & Penang); Ridership prediction, multi-feature analysis	Standard DL (multi-feature, non-hybrid); Limited operator features	✗ No explicit multi-scale TCN/window sweep; ✗ No MCO-included vs excluded evaluation; ✗ No graph/attention SOTA comparison	Closest Malaysia DL ridership work, yet standard architecture without hybrid multi-scale design or systematic paradigm benchmark

(S.K.B et al., 2024)	Urban mobility (multi-modal); Urban mobility patterns	GeoSpatial-Temporal LSTM (GT-LSTM); Multi-modal geospatial + temporal streams	Partial (geo-temporal LSTM, not multi-scale TCN blocks); ✗ No documented structural-break / MCO protocol; ✗ GT-LSTM vs traditional baselines only	Shows multi-modal DL gains but not hybrid graph–attention fusion or anomaly-segmented benchmarking
(Sobrie et al., 2023)	European railways; Real-time delay prediction	DL (LSTM encoder–decoder); Operational delay features	✗ Single-scale sequential encoding; ✗ No extreme-anomaly segmentation; ✗ Single architecture vs rule-based systems	Demonstrates DL superiority for transit prediction but single-paradigm, non-Malaysia, no ridership–MCO analysis
(Yin et al., 2022)	Survey (traffic prediction); Traffic flow / speed forecasting	Review: CNN, RNN, GCN, attention; Discusses multi-source conceptually	Discusses multi-scale conceptually; Notes distribution shift; no MCO case study; Compares families conceptually, not empirically on one dataset	Establishes DL over traditional models but does not operationalise hybrid multi-scale ridership forecasting under anomalies

(Jiang et al., 2021)	Survey + traffic benchmarks; Urban traffic prediction	Review + benchmark of DL traffic models; Multi-source in benchmark datasets	Limited multi-scale discussion; ✗ No pandemic-era ridership anomaly protocol; Partial (traffic DL families, not transit ridership paradigms)	Provides evaluation-metric framework (MAE, RMSE, MAPE) but not Malaysia transit or MCO-segmented multi-paradigm comparison
(Mystakidis et al., 2025)	Survey (traffic congestion); Congestion prediction	Review of emerging DL techniques; Multi-source discussed at survey level	Emerging multi-scale methods noted; ✗ No empirical MCO / ridership anomaly study; ✗ Survey only	Catalogues SOTA techniques but lacks Malaysia ridership empirical benchmark across sequential, graph, and attention models
(Maria-Arribas et al., 2026)	Traffic (graph-augmented dataset); Spatiotemporal traffic analysis / ML	Graph-augmented traffic dataset for DL; Graph + traffic sensor streams	Dataset supports multiple scales; Documents extreme mobility anomalies (85–90% drops); ✗ Dataset paper, not ridership forecaster benchmark	Confirms anomaly severity but does not quantify distortion of graph/attention SOTAs on Malaysian national ridership

(Zhao & Lan, 2025)	USA (highway reliability); Travel-time reliability modelling	Statistical / reliability modelling; Highway route data	✗ Not multi-scale DL forecaster; Addresses non-typical travel patterns; ✗ Not multi-architecture DL benchmark	Cited for confounding non-typical patterns; does not benchmark Malaysian transit DL paradigms under MCO segmentation

Note. ✓ = explicitly addressed in the cited study.

✗ = not addressed

Surveys provide methodological context; empirical ridership gaps are concentrated in (Farahmand et al., 2023) and (Amir et al., 2025).

CHAPTER 3

METHODOLOGY

3.1 Data Collection and Understanding

This study integrates eight heterogeneous data sources, daily ridership, GTFS network supply, GADM administrative boundaries, retail fuel prices, public and school holidays, state-level rainfall, gridded population density, and OSM point-of-interest counts, into a single spatiotemporal forecasting dataset. Per-source coverage, content, and usage are summarised in Table 3.1.

3.1.1 Primary Transit and Network Data

Daily ridership counts for 12 Malaysian service lines (LRT, MRT, Monorail, KTM, and RapidBus) and static GTFS data were obtained from data.gov.my. Combining GTFS with spatial layers supports granular assessment of network efficiency and accessibility (Hassan et al., 2025). Real-time GTFS streams are excluded because the portal provides only vehicle positions as of early 2026; deriving metrics from raw coordinates would degrade inference efficiency.

3.1.2 Administrative Geography and Spatial Connectivity

GADM level-1 boundaries for Malaysia were standardised to WGS-84 to construct binary and border-length-weighted adjacency matrices. Graph-based spatial structures capture regional heterogeneity in transit demand (Jin et al., 2022).

3.1.3 Economic and Temporal Indicators

Weekly retail fuel prices (RON95, RON97, diesel) from data.gov.my were forward-filled to a daily index; Malaysian public and school holidays (2019–2026) were extracted from timeanddate.com. A significant positive correlation between fuel price increases and public transit ridership in developing countries (Ahmad et al., 2024).

3.1.4 Environmental and Geospatial Features

Gridded population density (worldpop.org), 7-category POI counts within a 500 m stop radius (OSM Overpass API), and sub-national daily rainfall (data.gov.my) were integrated as environmental covariates. Non-linear ridership relationships require joint semantic and spatial encoding (Shi et al., 2024).

3.1.5 Data Segmentation and MCO Considerations

The dataset (1 January 2019 – 31 December 2025) was split into MCO-inclusive and MCO-exclusive configurations to measure how atypical operational periods affect model generalisation. Table 3.1 summarises all eight sources.

Table 3.1

Independent Raw Data Sources: Coverage, Content, and Usage in This Study

Source (Coverage)	Content	Usage
Ridership (2019-01-01 – present)	Daily passenger counts for 12 transit services + total	Primary forecast target; provides the 13 output variables (12 service lines + total)

Fuel Price (2017-03-30 – present)	Weekly retail prices for 6 fuel types (level + weekly change)	Demand driver; capturing modal substitution behaviour
GTFS Static (Snapshot)	Standard GTFS files for Rapid Rail KL, RapidBus KL, RapidBus Penang, KTMB	Network topology; stop counts, route density and travel-time edges quantify transit supply
Rainfall (2019-01-01 – present)	Daily precipitation at state level; 9 metrics including accumulation and anomaly indices	Weather covariate; state-level daily precipitation captures external signals
GADM (2020)	GeoJSON polygons for 16 Malaysian states (Level 1 admin divisions)	Spatial structure; used to derive inter-state adjacency weights
POIs (Snapshot)	OSM POI extract; 7 categories (transport, food, retail, education, healthcare, leisure, other)	Land-use context within a 500m catchment of each stop
Population Density (2020)	~1km^2 gridded raster of population density across Malaysia	Spatial context; static proxy for transit catchment potential
Holidays (2019 – 2026)	Date-range calendar of public holiday and academic break periods	Demand modifier; lead/lag features capture anticipation dips and post-holiday return surges

3.2 Feature Engineering

3.2.1 Data Cleaning

Eight independent preprocessing scripts transform heterogeneous raw formats into a standardised daily matrix (Table 3.2). Percent-change fuel features alongside level series capture short-run economic responses (Pour et al., 2026); cyclic sine/cosine encoding avoids ordinal discontinuities at calendar boundaries (Casolaro et al., 2023).

$$\mathrm{dow\_sin} = \sin\!\left(\frac{2\pi \cdot \mathrm{dow}}{7}\right), \quad \mathrm{dow\_cos} = \cos\!\left(\frac{2\pi \cdot \mathrm{dow}}{7}\right)$$

$$\mathrm{month\_sin} = \sin\!\left(\frac{2\pi \cdot \mathrm{month}}{12}\right), \quad \mathrm{month\_cos} = \cos\!\left(\frac{2\pi \cdot \mathrm{month}}{12}\right)$$

Table 3.2

Primary Cleaning Operations Applied to Each Sources

Source	Cleaning Operations
Ridership	Date parsing, MCO binary flag, pre-/post-launch null handling, total_ridership derivation
Fuel price	Daily forward-fill reindex, percent-change derivation, back-fill of leading nulls
Holidays	Expansion to daily rows, lead/lag proximity features, cyclic DOW/month encoding
Rainfall	Version filter (final only), wide pivot (date * state), 7-day linear interpolation, bfill/ffill
GTFS	Referential integrity verification, coordinate validation, directed edge extraction, static scalars
OSM POI	Element deduplication, 7-category taxonomy, 500m catchment counts, per-stop log transforms

Population Density	Bounding-box filter, log1p transform, state aggregation, national median scalars
GADM Boundaries	Property cleaning, centroid extraction, shared-border adjacency matrix

3.2.2 Feature Alignment

All variables were aligned to consecutive daily indexes for both research periods using the merge and null-handling rules in Table 3.3. Autoregressive lags (7, 14, 28 days) were computed from the full historical ridership series before merging (Wu et al., 2023; Yusuf et al., 2025), yielding a 79-feature matrix grouped as in Table 3.4.

Table 3.3

Merge Strategy and Null Handling

Source	Join Method	Null Handling
Ridership (Targets)	reindex	Pre-launch nulls retained; filled to 0.0 in sequence assembly
Fuel Price	reindex + ffill	No gaps after forward-fill
Holiday / Cyclical	reindex + fillna(0)	Non-holiday days default to 0
Rainfall	reindex + interpolate(limit=7) + bfill/ffill	Contiguous after interpolation
Static Features	Scalar broadcast	No nulls across population density, GTFS, POI, or GADM streams

Table 3.4

79 Features Across 5 Semantic Groups

Group	Count	Description
Targets	13	12 service-line ridership counts + total_ridership
Temporal	16	Holiday flags, lead/lag proximity days, cyclic DOW/month, year, day_of_year, is_weekend
External	30	Fuel price levels and percent-changes (15) + per-state rainfall (15)
Lag	3	ridership_lag_7, ridership_lag_14, and ridership_lag_28
Static	17	Population density, GTFS network counts, OSM POI means, GADM boundary stats
TOTAL	79	Finalized aligned multivariate input matrix

3.2.3 Sequence Building

The aligned matrix was converted to sliding-window tensors (T_{in} look-back, T_{out} = 7-day horizon) with a 70%/15%/15% chronological train/validation/test split. MinMax scaling was fit on training data only to prevent leakage (Casolaro et al., 2023).

3.3 Explanatory Data Analysis

3.3.1 MCO Collapse and Unequal Recovery

The MCO enforced between 18 March 2020 and 31 December 2021 is the single largest disturbance in the series, spanning roughly a quarter of the 2019 – 2025 observations. Figure 3.1 plots total daily transit volume across the full study window, where volume falls abruptly by 85 – 90% at the onset of the MCO and the six detected structural shifts delimit four behavioural phases, which Figure 3.2 aggregates into the three reporting phrases used throughout this study and shows that the return was uneven: rail-heavy services recovered to a lower fraction of their own baseline than bus services, reflecting rigid network topology against route-level flexibility (Lv et al., 2021).

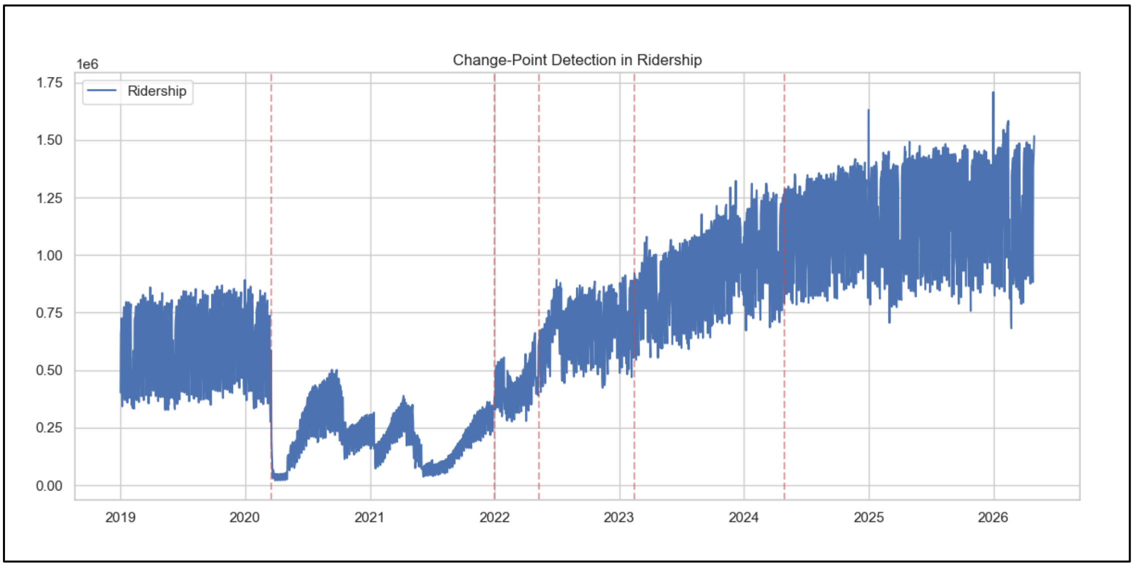

Figure 3.1 - Total Daily Transit Volume from 2019 to mid-2026, Featuring 6 Vertical Red Dashed Lines Pinpointing Specific Structural Shifts/Transition Points over Time

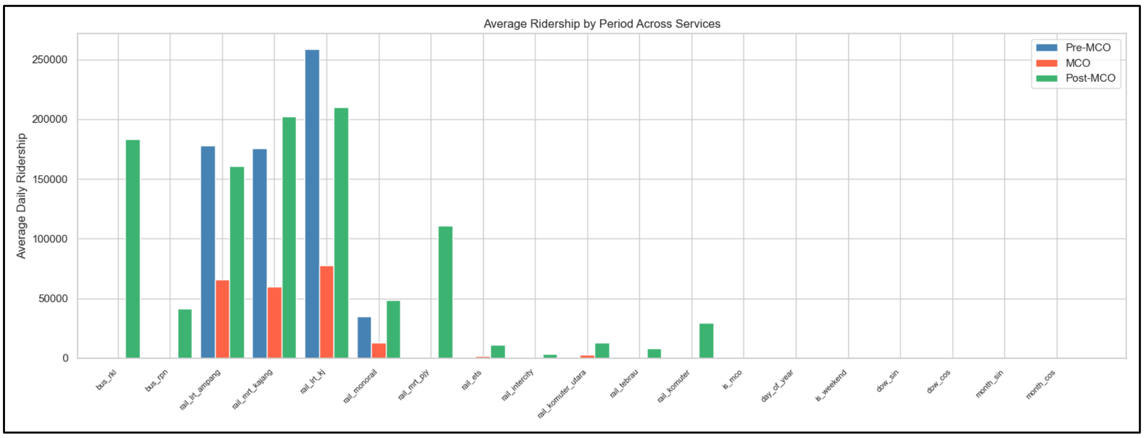

Figure 3.2 - Average Daily Usage Numbers Across Various Rail and Bus Lines Segmented into 3 Distinct Phases: Pre-MCO (blue), MCO (orange), and Post-MCO (green)

The asymmetry visible in these two figures is best explained by which travel purposes each mode carries and how substitutable those purposes were. Based on Figure 3.1, the structural shifts marked by the red dashed lines are not evenly spaced: the collapse is near-instantaneous while the return is staged across several transitions, indicating that the recovery was governed by the sequential release of activities rather than by a single policy date. Figure 3.2 resolves the same period by service, and comparing the green post-MCO bars against the blue pre-MCO bars line by line shows that no two services returned to the same fraction of their own baseline.

These gaps are not random. Urban rail is built around fixed corridors feeding the Kuala Lumpur central business district, and precisely that segment, office commuters, retained the option of remote work, so a portion of its demand was removed rather than displaced. Bus services carry a larger share of trips that cannot be performed from home and therefore returned earlier, whereas inter-city rail depends on discretionary leisure and business travel, the last category to be released. MRT Kajang stands out in Figure 3.2 because it was still climbing its post-launch maturation curve when the MCO struck, so latent network growth compounded with recovery. The practical consequence is that the mapping from calendar and weather inputs to ridership is not the same function before and after the break; both the mean level and the variance shift (Lv et al., 2021). This motivates the dual mco/nomco dataset construction in Chapter 3.2 and the regime-aware components introduced in Chapter 3.5 (Amir et al., 2025).

3.3.2 Northeast Monsoon Rainfall Pattern

Malaysian rainfall is both seasonal and strongly regional. Figure 3.3 compares the total counts of wet day (≥10 mm), very wet days (≥30 mm) and extreme days (≥50 mm) across states, with Terengganu recording the highest count at every threshold. Figure 3.4 resolves the same measure by month and by state, showing that extreme-rainfall frequency peaks between November and January in line with the Northeast Monsoon, and that this peak is concentrated in the east-coast states while west-coast frequencies remain comparatively low throughout the year. Figure 3.5 presents the corresponding national rainfall trend and its monthly anomalies. Aggregating the fifteen-state series to a single national figure therefore conceals the regional variation that drives mode-specific travel choices (Ngo & Bashar, 2024).

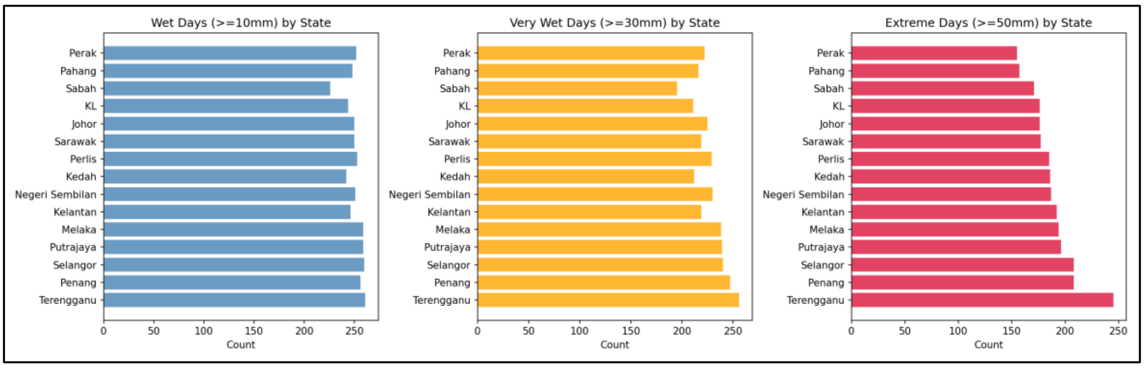

Figure 3.3 - Comparison of Total Counts of Wet Days (≥10mm), Very Wet Days (≥30mm), and Extreme Days (≥50mm) Across Various States with Terengganu Consistently Showing the Highest Counts

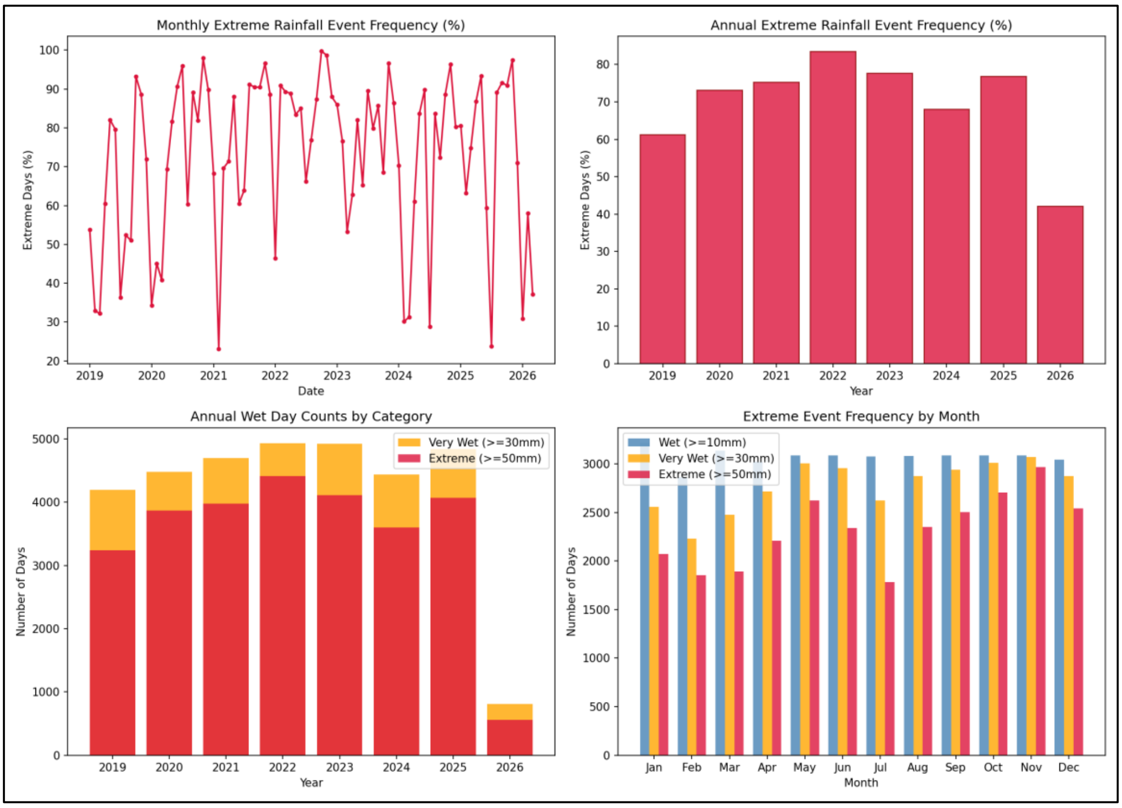

Figure 3.4 - Extreme Rainfall Frequency by Month and State, 2019–2026

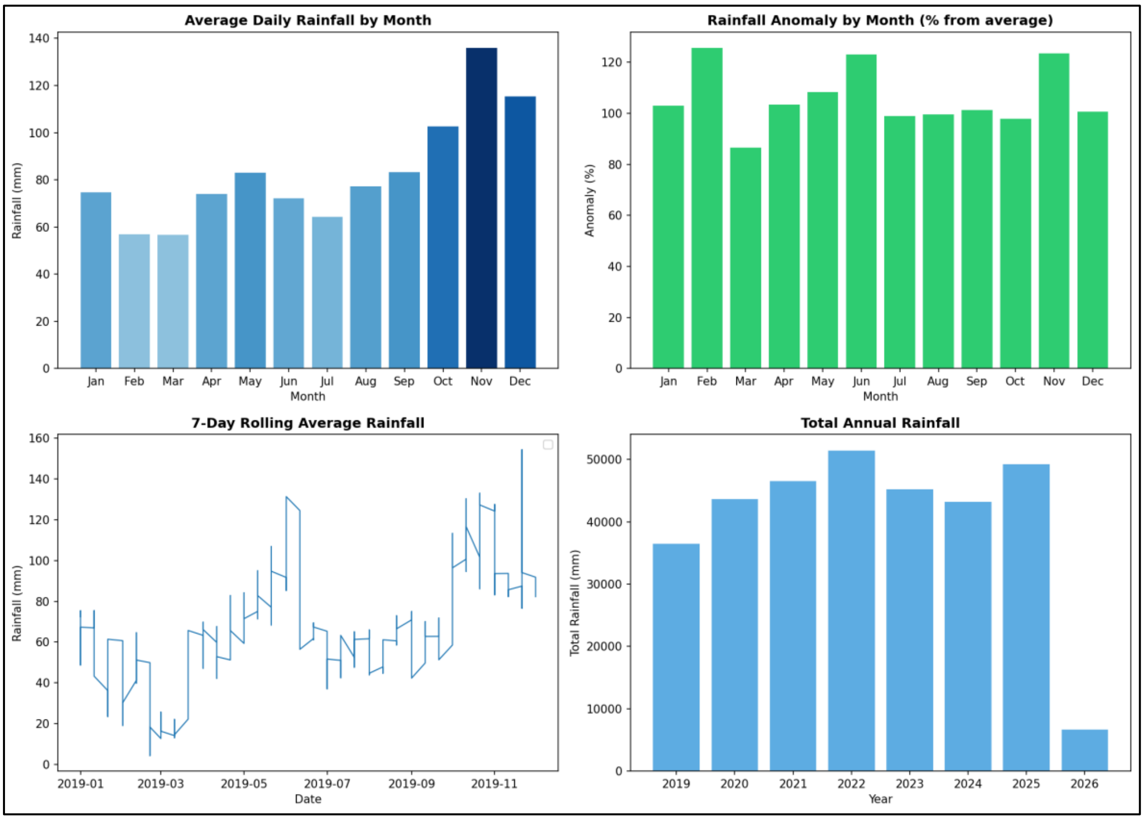

Figure 3.5 - National Rainfall Trends and Monthly Anomalies, 2019–2026

The reason a national rainfall average is misleading here is geographic rather than statistical. Based on Figure 3.3, the east-coast states, Terengganu highest among them, followed by Kelantan and Pahang, record several times the extreme-day count of Selangor and Kuala Lumpur, and the gap widens as the threshold rises from wet to very wet to extreme, so the disparity is concentrated precisely in the heaviest events. Figure 3.4 shows that this is a single joint pattern rather than two independent ones: The November-January band is confined almost entirely to the east-coast rows of the month-by-state grid. The physical cause is the Titiwangsa range, which shelters the western states from the Northeast Monsoon, so one synoptic system can produce flooding in Kota Bharu on a day the Klang Valley stays dry. Figure 3.5 demonstrates the analytical consequence directly, once the same data is collapsed to a national series, the monthly anomalies flatten into a mild, near-symmetric oscillation and the extremes visible in Figures 3.3 and 3.4 disappears.

Because the great majority of national ridership is generated on the west coast, a national mean would be pulled by east-coast extremes that carry almost no ridership exposure, attenuating the true weather-demand relationship and potentially reversing its sign. Retaining all fifteen state-level rainfall columns instead lets the model learn a separate weight per state and, in effect, discover for itself which states carry demand (Ngo & Bashar, 2024). The same reasoning sets the seven-day cap on interpolations: a monsoon event of the kind visible in Figure 3.4 lasts a few days, so seven days bridges one event without merging two into a fabricated wet spell (Jiang & Cai, 2023).

3.3.3 Weather-Induced Ridership Surge under Heavy Rainfall

The relationship between rainfall and ridership is not monotonic. Figure 3.6 contrasts ridership under heavy-rain conditions against normal conditions and shows that heavy-rain days do not carry the reduced ridership a simple discouragement effect would predict. Figure 3.7 plots monthly mean daily rainfall against total ridership together with the monthly correlation coefficients and their associated p-values, where the coefficients vary in both magnitude and sig across the calendar. Figure 3.8 pools every day into a single rainfall-ridership scatter and wet-day comparison, returning an overall correlation of r = 0.14. Read together, these results indicate that light rains slightly suppress travel while rainfall above roughly the 75th percentile is associated with increased transit ridership, consistent with modal substitution during hazardous driving conditions (Huang et al., 2022).

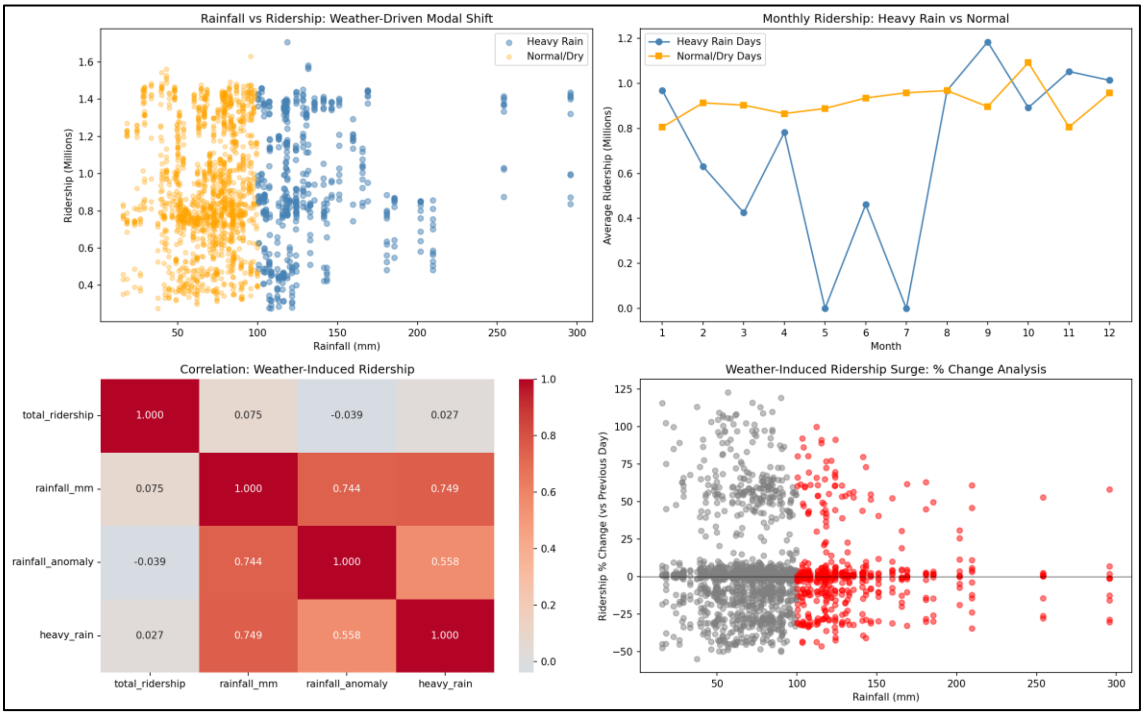

Figure 3.6 - Ridership Response to Heavy-Rain Versus Normal Conditions

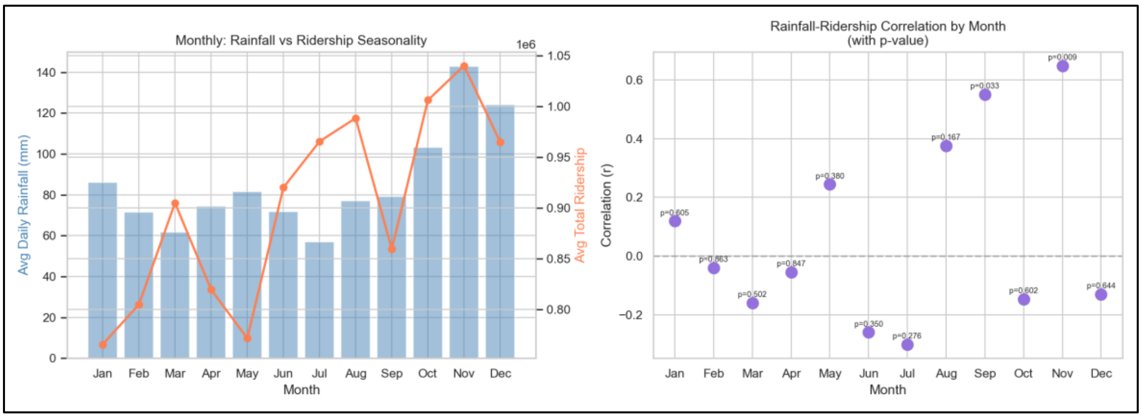

Figure 3.7 - Monthly Averages for Daily Rainfall against Total Ridership with Monthly Breakdown of Correlation Coefficients, r, with Corresponding Statistical Significance (p-values)

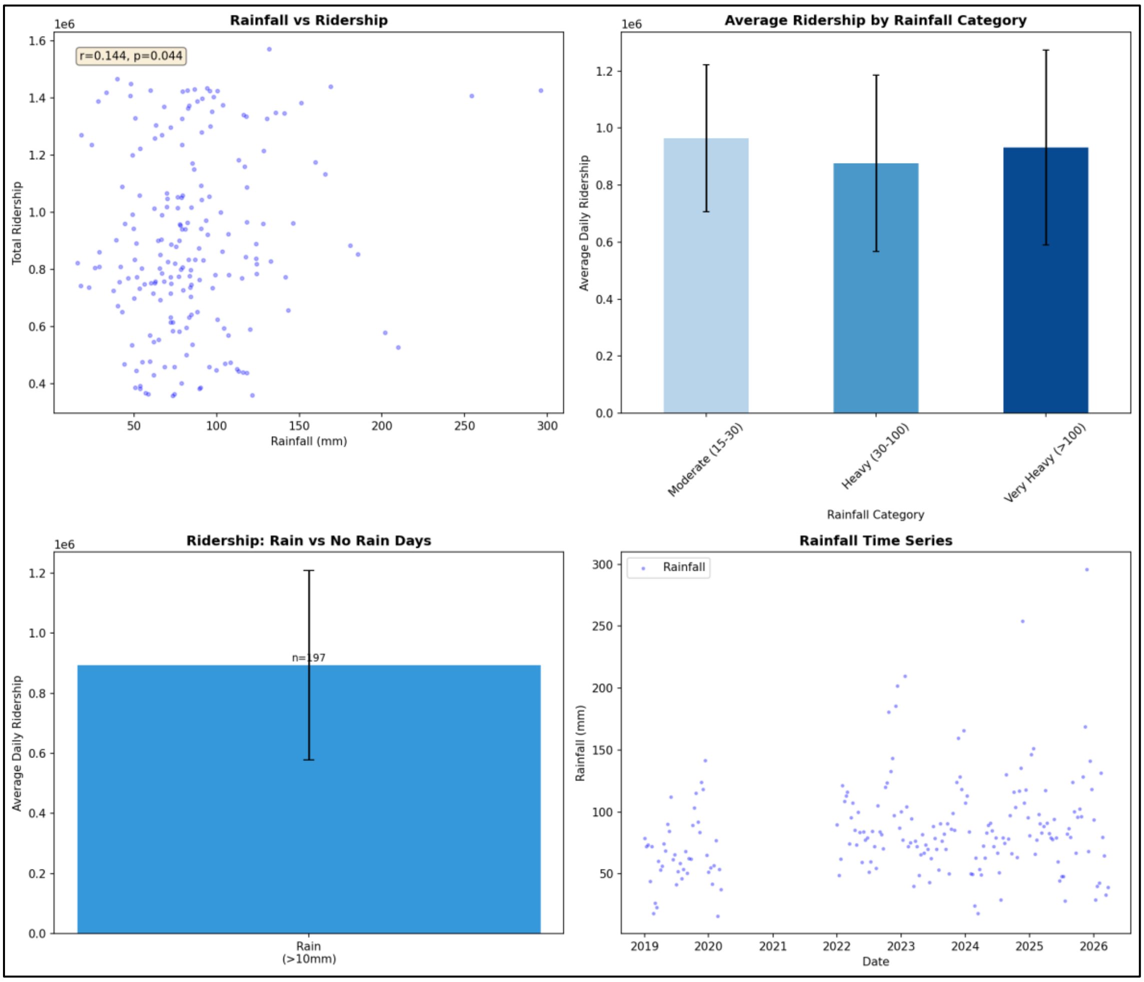

Figure 3.8 - Rainfall–Ridership Scatter and Wet-Day Ridership Comparison (r = 0.14)

This non-monotonic response arises because two opposing mechanisms dominate at different rainfall intensities, and the three figures are best read as three views of that same duality. Based on Figure 3.6, no single monotone rain-discourages-travel effect can account for the heavy-rain bar sitting where it does. The sign reversal across months in Figure 3.7 then locates the discontinuity in the calendar: the relationship is not uniformly weak but switches direction, which is the signature of two processes with different intensity thresholds rather than one noisy process.

At low intensity the suppression effect prevails, discretionary trips are deferred and the disutility of walking leg to a stop or station rises. Above roughly the 75th percentile a substitution effect takes over, as surface roads flood and driving becomes dangerous, so car users transfer onto grade-separated rail that is largely insulated from surface conditions; buses gain far less because they share the same flooded carriageway and still require exposed kerbside wating. Figure 3.8 quantifies what happens when the two mechanisms are averaged: the pooled correlation of r = 0.14 is weak not because rainfall is unimportant but because opposing effects cancel within a single coefficient. Two design decisions flow. Rainfall is retained as a continuous value rather than a binary wet-day flag, since a binary flag can express only one sign of effect (Huang et al., 2022). Second, all twelve services are forecast alongside the total, so a rainfall response that is positive for rail and negative for bus need not be collapsed into one national parameter (Farahmand et al., 2023).

3.3.4 Fuel Price Elasticity and Modal Substitution

Retail fuel prices enter the dataset as both levels and changes, and the two behave differently. Figure 3.9 plots RON95, RON97 and diesel in MYR per litre alongside the weekly price changes for RON95 and diesel, showing that prices are revised on a weekly cycle and held constant between announcements. Figure 3.10 relates those price levels to average daily ridership, where the post-MCO Pearson correlation between RON95 and total ridership is approximately +0.35 and rises non-linearly above RM2.20 per litre. Figure 3.11 reports the correlation between changes in RON95 and diesel prices and subsequent changes in ridership at lags of zero to seven days, with the response peaking at one to three days. Separating the level series from the price adjustments therefore prevents a model from confusing multi-year trends with sudden policy shocks (Chevance et al., 2024).

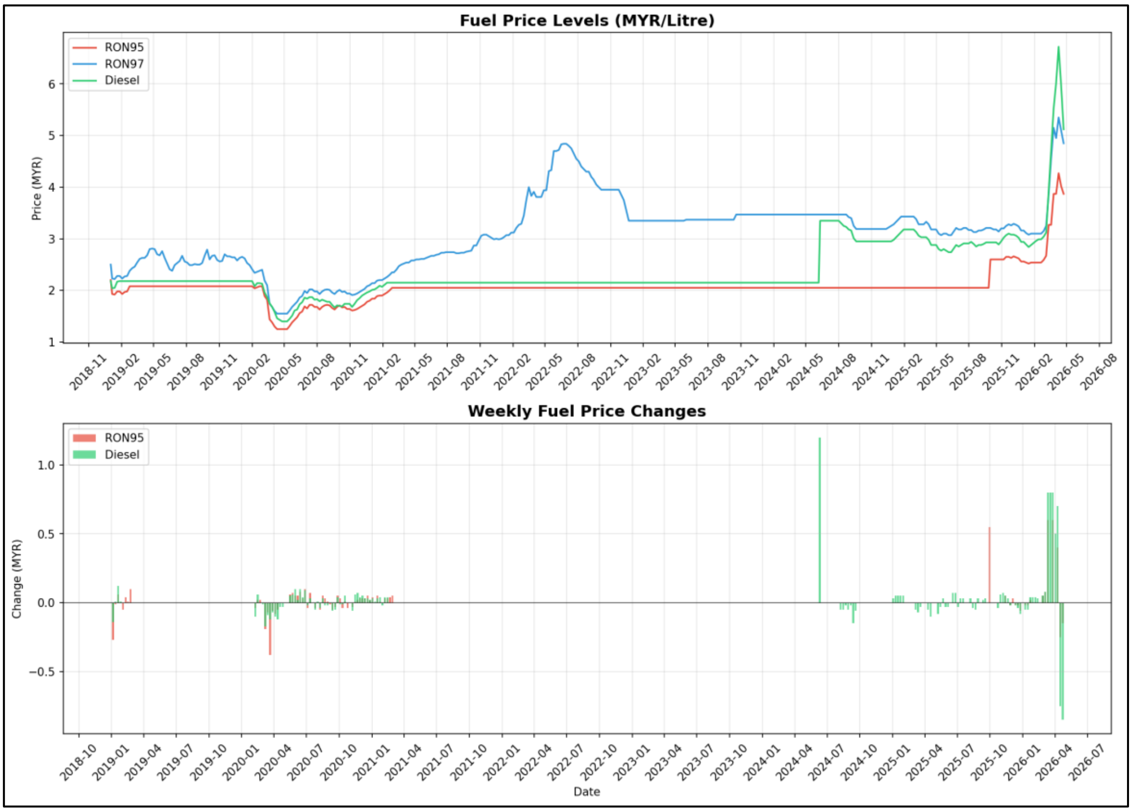

Figure 3.9 - Fuel Price Levels (RON95, RON97, Diesel) in MYR/Litre (top) and Weekly Price Changes for RON95 and Diesel (bottom), both from late 2018 to mid-2026

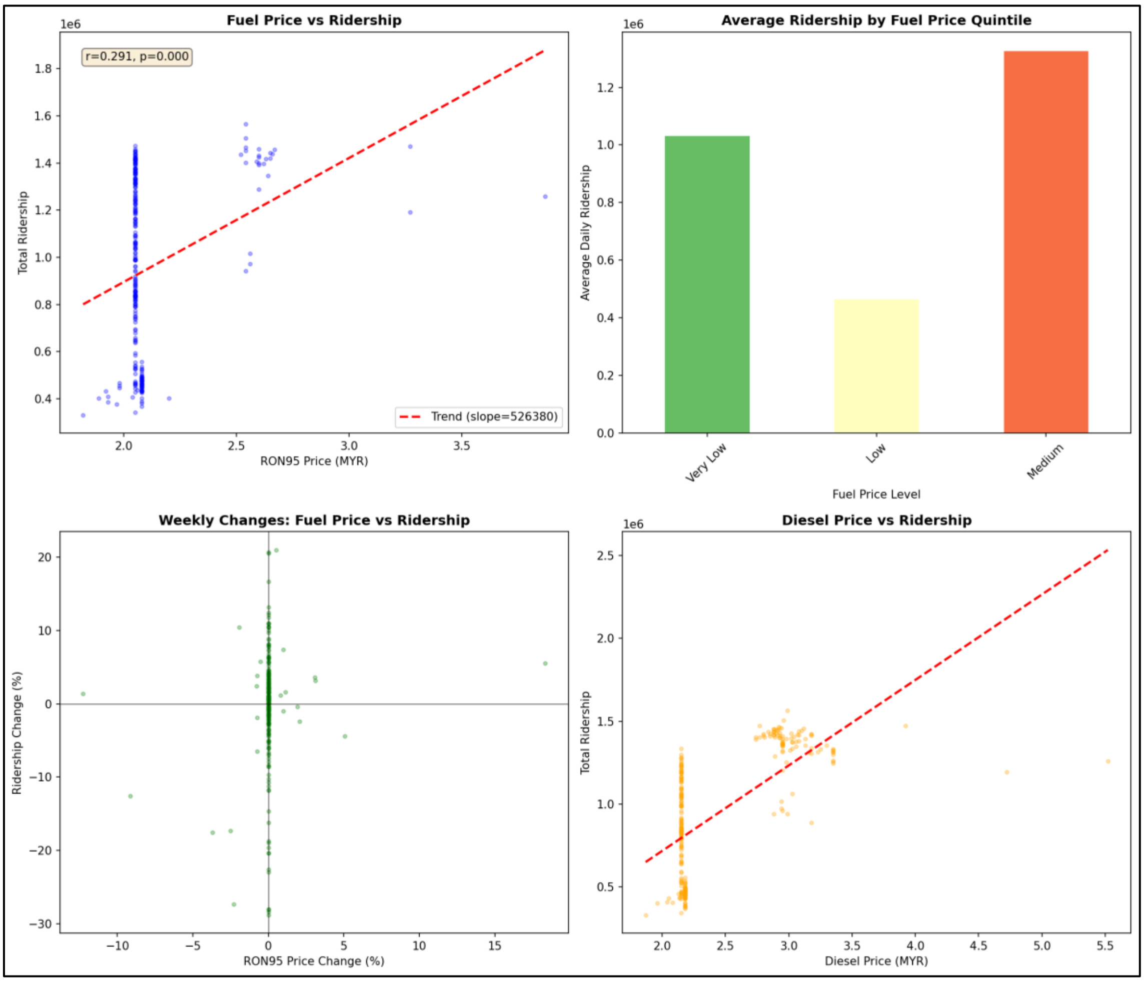

Figure 3.10 - Fuel Price Levels Versus Average Daily Ridership

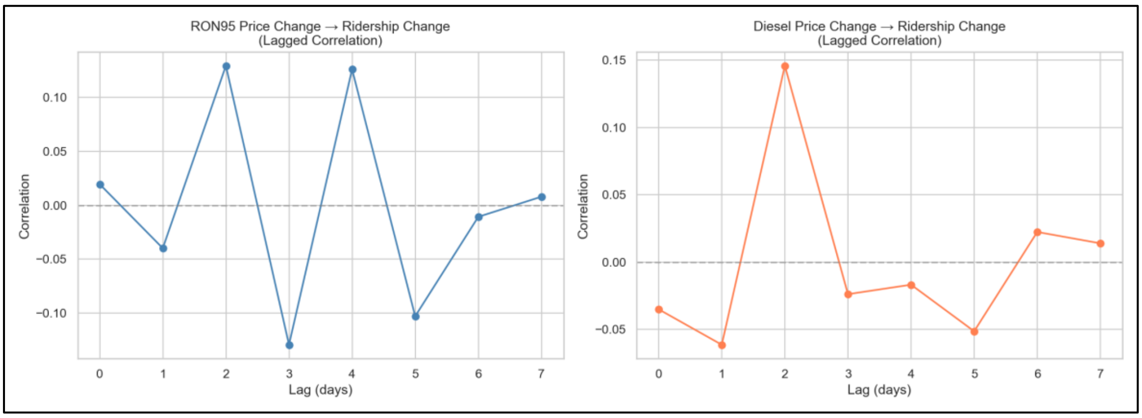

Figure 3.11 - Lagged correlation (from 0 to 7 days) between Changes in Fuel Prices (RON95 and Diesel) and Subsequent Changes in Transit Ridership

The positive level-on-level correlation of about +0.35 should not be read as an elasticity and understanding why explains the feature design. Based on the upper panel of Figure 3.9, the weekly announcement cycle makes the price a staircase rather than a continuously varying quantity, and a staircase has an autocorrelation that decays almost as slowly as a unit-root process. Two series that both behave in this way will correlate strongly with each other whether they are casually related. That is precisely the risk in Figure 3.10, where the apparent positive association is largely a shared upward drift, both the administrated price path and ridership rose across the post-MCO recovery, so a level-on-level correlation captures common trend rather than substitution.

The genuine behavioural mechanism is threshold-and-lag: motorists do not re-optimise mode choice on a single price notice, because the cost is amortised across a full tank and becomes salient only at refuelling. Figure 3.11 is the direct evidence for this. The lagged correlation is near zero at day 0 and peaks one to three days later, which is the signature of a delayed behavioural response rather than a contemporaneous accounting relationship (Chevance et al., 2024). Signed weekly percentage and absolute change series, the quantity plotted in the lower panel of Figure 3.9, are therefore added alongside the levels, isolating the moment a price moves. The subsidised RON95 variants introduced from mid-2023 are administratively frozen and carry almost no variance, which is why they later fall out under SHAP-guided reduction (Pour et al., 2026).

3.3.5 Holiday Anticipation and Return-Surge Effect

Holiday effects extend well beyond the holiday itself. Figure 3.12 sets the seasonal and weekly ridership profile against the distribution of public holidays across the study period. Figure 3.13 groups the observed dip by holiday duration and shows that short holidays produce the largest reduction. Figure 3.14 separates academic from public holidays, with public holidays producing sharp falls in both mean volume and daily percentage change while academic breaks shift the level more gradually over a longer span. Figure 3.15 isolates the holiday window itself, where ridership falls by approximately 18% one to two days before a public holiday and rises sharply on the first working day afterwards. Proximity features expressing the number of days to and from each holiday type allow a model to learn these demand gradients directly.

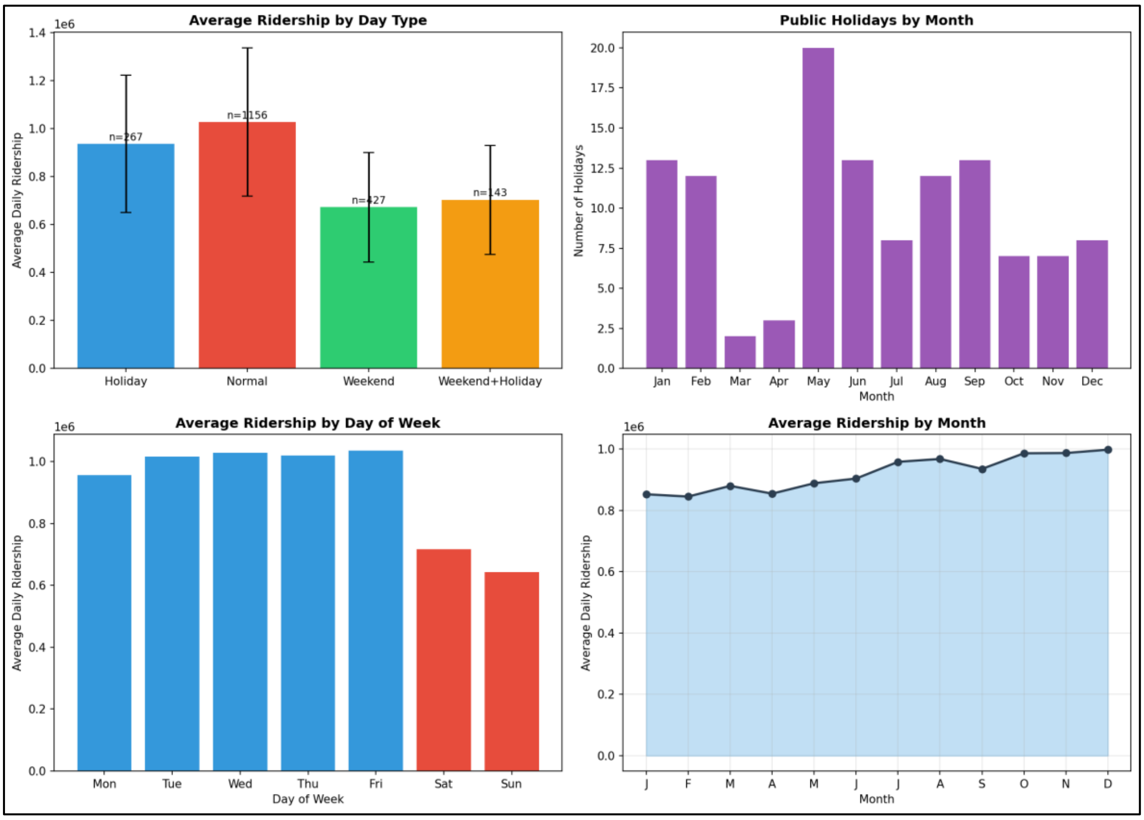

Figure 3.12 - Seasonal and Weekly Ridership with Public-Holiday Distribution

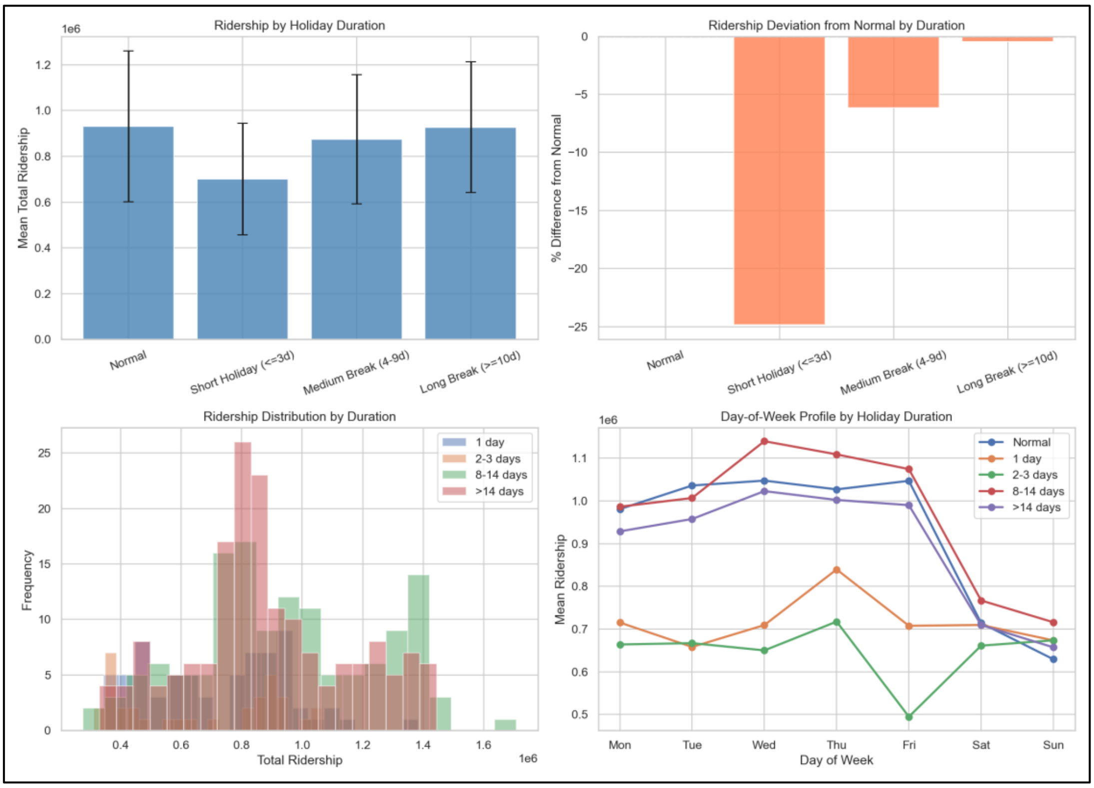

Figure 3.13 - Ridership Dip by Holiday Duration (Short Holidays Worst)

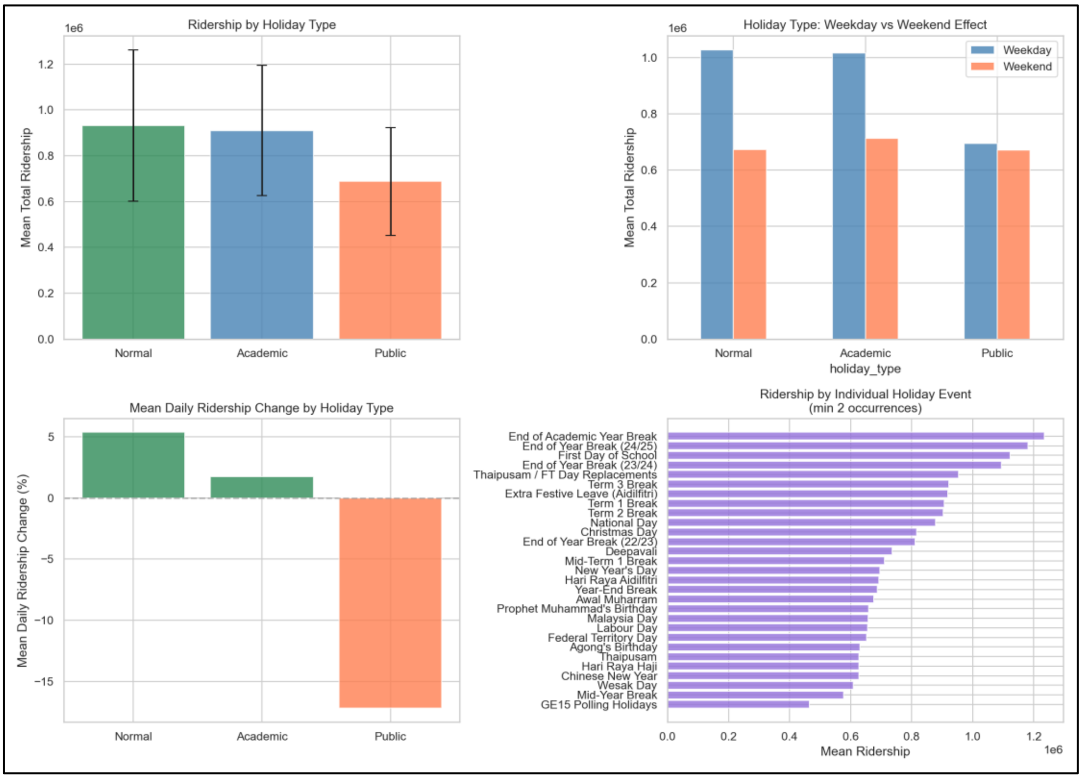

Figure 3.14 - Ridership Trends between Academic and Public Holidays highlighting Public Holidays See Sharp Drops in Both Overall Mean Volume and Daily Percentage Change with Breakdown of Mean Ridership Across Dozens of Individual Malaysian Calendar Events

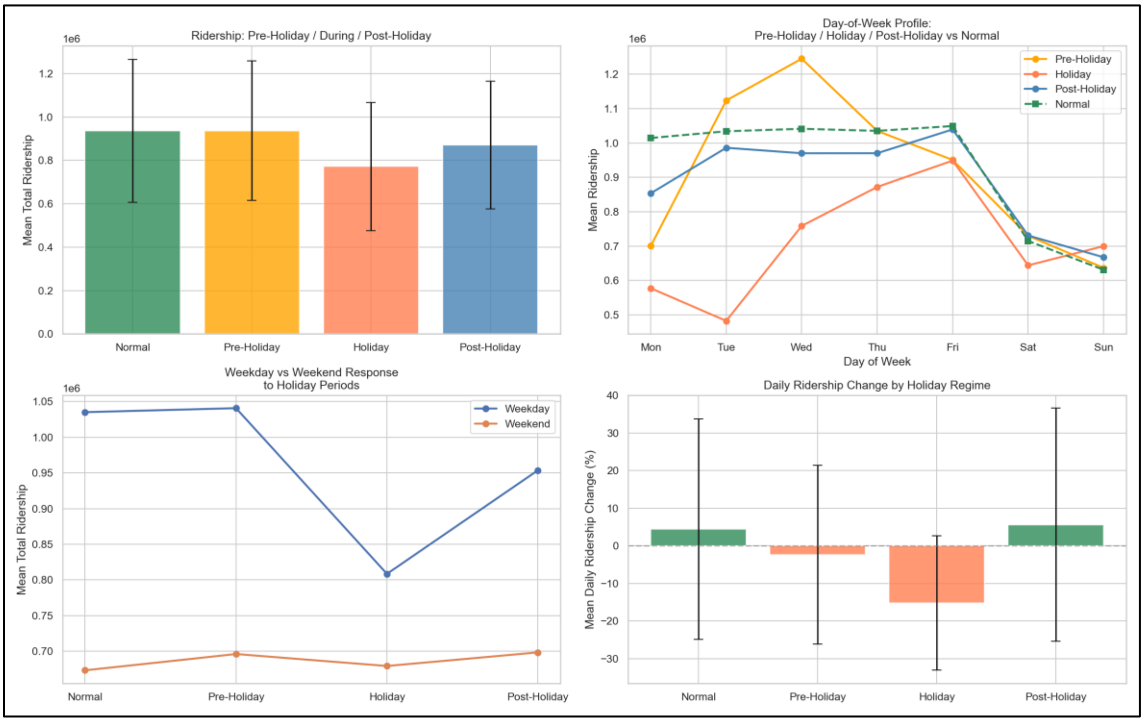

Figure 3.15 - Holiday-Period Demand: Pre-Dip and Post-Surge Patterns

The pre-holiday dip and post-holiday spike are two halves of one displacement rather than two separate events, which is why a point-in-time flag cannot represent them. Based on Figure 3.15, the two halves are not symmetric, and the asymmetry is informative: departure can be brought forward at the traveller’s discretion whereas employment resumes on a fixed date, so the outflow spreads across two days while the return compresses into one. The underlying driver is the balik kampung practice, major festivals trigger a physical migration out of the Klang Valley, and commuters take leave on the bracketing days to extend the break, so the urban commuting base drains before the holiday and refills after it. Figure 3.13 corroborates this from another direction.

That short holidays produce the deepest dip is counter-intuitive if the effect were simply proportional to days away but follows naturally from displacement: a short break forces the entire outflow and return into a narrow window, whereas a longer break allows both to spread. The contrast in Figure 3.14 has the same origin, a public holiday relocates the population briefly and sharply, while an academic break alters the composition of demand for weeks without any comparable daily discontinuity. Two encoding decisions follow. A binary is_public_holiday flag would capture only the trough visible in Figure 3.12 and force the model to treat the flanking days as noise, so four directional counters, days until and days since each holiday type, supply a monotone ramp on either side (Wu et al., 2023). The two holiday types are kept as separate flags for the same reason: merging a sharp one-day commuter drop with a multi-week academic break would weaken both signals (Rahmani et al., 2025).

3.3.6 Lag Dominance and Static Near-Zero Signal

The predictor set is dominated by a small number of autoregressive terms. Figure 3.16 pairs a 6 x 6 correlation matrix of the key demand-forecasting inputs with the standardised coefficients of the dominant predictors of ridership, where the ridership lags at 7, 14 and 28 days carry the largest weights; a principal component analysis of the same correlation matrix attributes approximately 72% of the total variance to the first component, which loads primarily on those lags. Figure 3.17 extends the view to the full 79 x 79 feature correlation matrix, where the static network and demographic features return near-zero linear correlation with the forecast targets and with one another. Their contribution, if any, is therefore not linear and cannot be assessed by correlation alone (Wu et al., 2023).

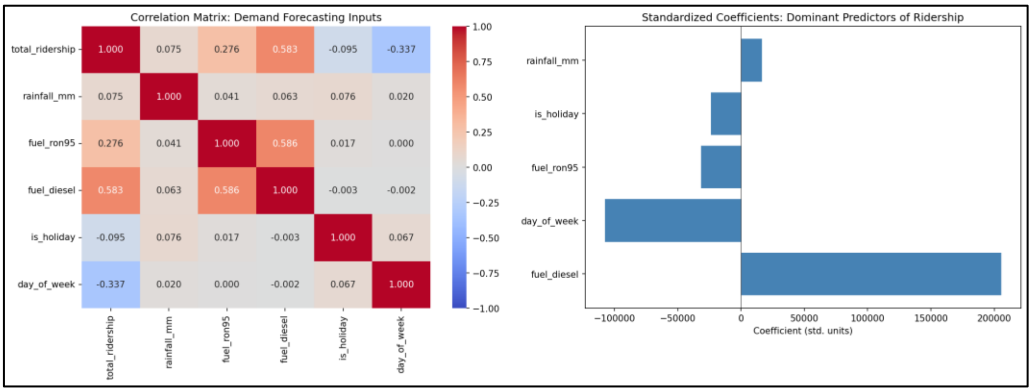

Figure 3.16 - Detailed 6 x 6 Correlation Matrix of Key Demand Forecasting Inputs (left) with Standardized Coefficients of the Dominant Predictors of Ridership (right)

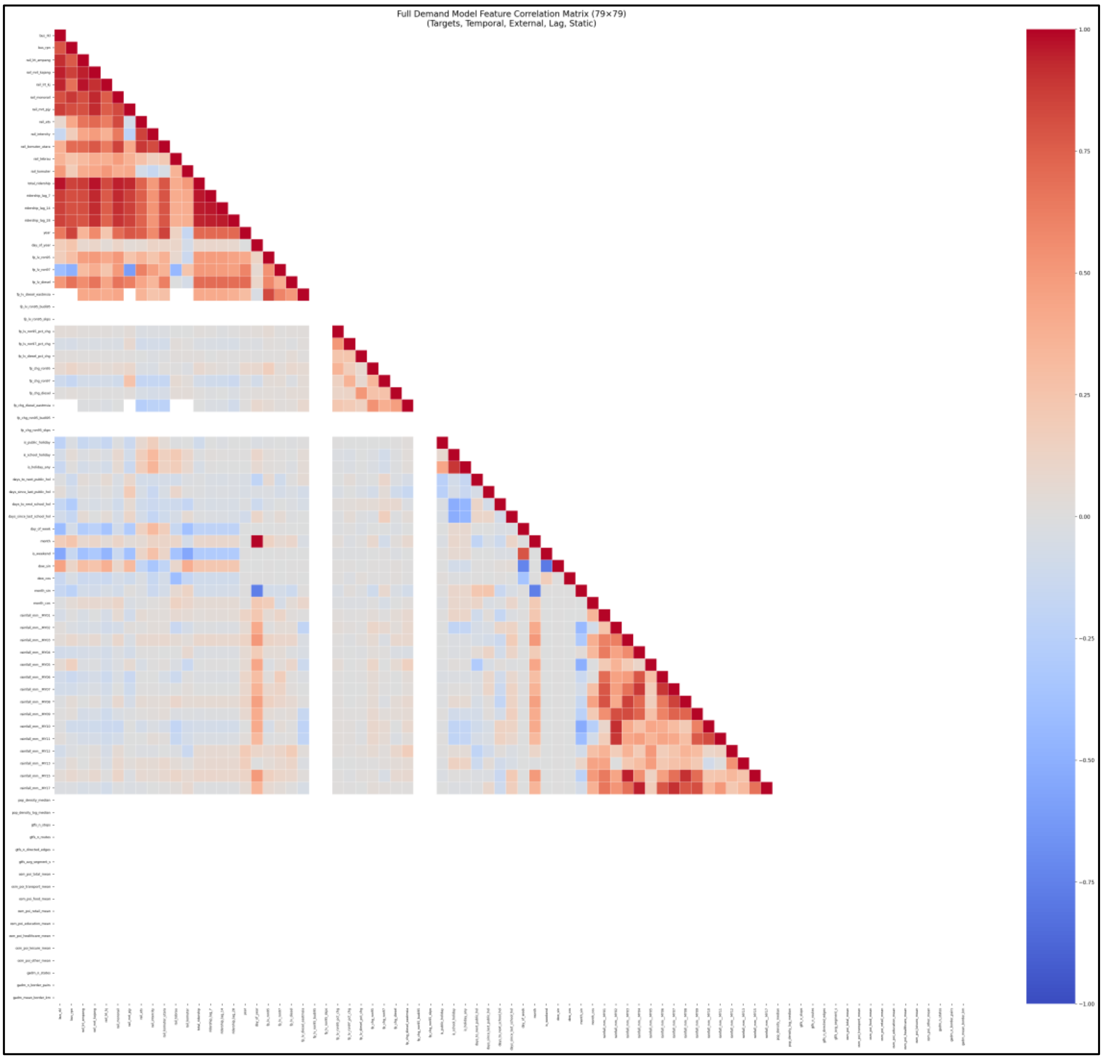

Figure 3.17 - Large-scale, Lower-triangular Heatmap visualizing Complex Feature Correlation Matrix (79 x 79) for Full Demand Model

These two findings have different causes and therefore call for different responses. Based on the right-hand panel of Figure 3.16, lag dominance reflects the nature of the demand process rather than a modelling artefact: public transit trips are generated by recurring commitments, work and school, that reproduce themselves on a weekly cycle, so the same weekday one week ago is a near-sufficient statistic for today under normal conditions. The left-hand panel makes the same point structurally, with the lag terms forming a tightly inter-correlated block distinct from the exogenous inputs. The informative part is where that baseline fails: residuals from a naïve seven-day-lag predictor are largest around public holidays and at the COVID regime boundary, which is precisely where the calendar, weather and fuel features earn their place.

The near-zero correlation of the static features has an entirely different explanation, and Figure 3.17 shows it as a visual signature, the static block appears as a uniformly pale region in the 79 x 79 heatmap, correlating with neither the targets nor the other predictors. Population counts, GTFS stop statistics, OSM points of interest and administrative areas are constant across the study window by construction, and a column with no temporal variance has essentially no linear correlation with anything that does vary, regardless of its causal importance. Such features can act only as fixed scaling that interact multiplicatively with the time-varying drivers, which linear screening cannot detect (Shi et al., 2024). The pale band in Figure 3.17 is therefore evidence of an inadequate measure, not of irrelevant features, which is why the reduction reported in Chapter 3.5 is driven by SHAP attribution rather than correlation ranking, and why all seventeen static columns are ultimately removed without loss of accuracy (Cui et al., 2025).

3.4 Baseline & Comparative Models

14 baseline deep-learning models form a progressive comparison chain that isolates each architectural contribution, from sequential encoding through graph propagation to series decomposition. Architecture diagrams, hyperparameters, and training configurations are documented in Appendix C.

3.4.1 Spatiotemporal Models

LSTM — Foundational sequential baseline establishing the performance floor (Barath et al., 2026). Gated memory captures daily and weekly commuter cycles without graph or attention inductive biases.

BiLSTM — Tests bidirectional encoding over the look-back window relative to LSTM (Krishnasamy et al., 2025). Relevant for mid-window holidays and disruptions whose ridership response follows within the observed window.

TPA-LSTM — Tests pattern-level attention over LSTM hidden states for multi-scale periodicity (Wei et al., 2023). Matched to weekday/weekend asymmetry and holiday suppression in Malaysian ridership.

CNN-LSTM — Tests 1-D CNN local feature extraction before recurrent encoding (Topilin et al., 2025). Three fusion modes (sequential, parallel, augmented) are evaluated per Appendix C.

CNN-BiLSTM — Combines CNN local extraction with bidirectional context (Zhuang & Cao, 2022). Most expressive LSTM-family model before explicit spatial stream decoupling.

ST-LSTM — Bridge model decoupling temporal (LSTM) and spatial (MLP) streams (Cui et al., 2025). Canonical reference for training loop and comparison logic across the codebase.

3.4.2 Graph Models

STGCN — Foundational graph baseline with fixed Pearson inter-feature adjacency and Chebyshev graph convolution (Deng, 2025). Transitions from implicit to explicit cross-feature relational encoding.

PDR-STGCN — Novel extension combining dynamic attention adjacency with periodicity-aware 2-channel input (Hu et al., 2026). Addresses static-graph and single-channel limitations of STGCN.

STSGCN — Fuses spatial and temporal graph operations in a single synchronous 3N×3N adjacency (Chen et al., 2023). Suited to simultaneous multi-feature shifts during holidays.

STFGNN — Dual graphs for cross-feature co-variation (spatial) and intra-window temporal-profile similarity (temporal), fused via a learnable gate (Chang et al., 2025).

MTGNN — End-to-end learned asymmetric adjacency from node embeddings (Wu et al., 2025). Tests whether task-specific graphs outperform Pearson correlation.

3.4.3 Attention Models

ASTGCN — Adds learnable spatial and temporal attention over the STGCN backbone (Cui et al., 2025). Adapts node and timestep weighting across normal weekdays versus holiday periods.

Autoformer — Series decomposition with FFT-based autocorrelation attention (Ma & Zhang, 2025). Tests periodic ridership as signal decomposition rather than graph propagation.

Informer — ProbSparse efficient transformer with generative multi-step decoder (Song et al., 2024). At $T_{in} = 14$, ProbSparse degenerates to near-full attention, ensuring fair comparison with Autoformer.

3.5 Hybrid Multi-Scale Temporal Spatio-Feature (HMT-TSF)

The term hybrid is used in this study in its established architectural sense: a model is hybrid when it integrates two or more distinct modelling paradigms, each carrying a different inductive bias, inside a single end-to-end trainable predictor. This is the sense in which the DL time-series literature already uses the term, and it is distinct from two weaker usages that are sometimes conflated with it, ensemble hybridity, which averages the outputs of separately trained models, and pre-processing hybridity, which chains a statistical decomposition to a neural predictor (Casolaro et al., 2023). Under this definition, hybridity is not a claim to have invented a new mathematical primitive. Every primitive used in HMT-TSF is prior art and is attributed as such in Chapter 3.5.2: reversible instance normalisation (Kim et al., 2023), dilated causal temporal convolution and multi-head self-attention (Jin et al., 2022), spectral graph convolution over a correlation-derived adjacency (Deng, 2025), and gradient-boosted residual correction. What is claimed is the composition, the specific selection of paradigms, the routing between them, and the gating that decides how their representations are combined.

HMT-TSF is hybrid along three axes simultaneously. It is paradigm-hybrid, because it draws its temporal encoder from the recurrent-convolutional family evaluated in Series 1, its relational encoder from the graph-based family in Series 2, and its global-context encoder from the attention-based family in Series 3, so that all three baseline paradigms are represented inside one network rather than compared beside it. It is representation-hybrid, because the global, local, relational and distributional views of the same input are not concatenated but combined through a learned squeeze-and-excitation gate that decides per sample how much each stream contributes. And it is learning-paradigm-hybrid, because the differentiable network is coupled with a non-differentiable gradient-boosted residual corrector that is applied only when it improves validation performance (Jiang et al., 2021). The design contribution therefore lies in the routing rather than the parts: the graph and regime streams read the instance-normalised raw feature tensor directly, deliberately bypassing the feature-group fusion and the temporal transformer, because those layers project the 53 feature channels into an abstract 128-dimensional space and destroy the feature identity that a features-as-nodes graph and a regime detector both depend on. This asymmetric tri-stream routing, illustrated in Figure 3.18, is what allows a single model to address all three limitations listed below at once, and it is the reason the three limitations are stated together rather than tackled by three separate models (Hu et al., 2026).

HMT-TSF is developed to address three key limitations identified across 14 baseline models:

- **No model captures all three types of structure simultaneously.** LSTM family models handle temporal sequences well but have no graph structure and no regime awareness; graph-based models propagate inter-feature correlations but are rigid under distributional shift, attention-based models decompose periodic signals effectively but cannot propagate domain-meaningful feature-to-feature messages.
- **No model explicitly handles the MCO regime shift.** All graph-based models use static. Pearson graphs computed on training data; the MCO pandemic caused a structural break in ridership patterns that invalidates these learned correlations.

Static normalisation strategies compound this problem by applying fixed-scale transformations across data from fundamentally different regimes.

- **No model uses domain-aware feature grouping.** All 14 baselines treat the 79 features as a flat vector, giving equal initial weight to structurally disparate sources.

Each limitation is answered by a named, separately abatable module rather than a general appeal to model capacity, and Figure 3.18 labels the three corresponding bands explicitly. The first limitation that no single baseline captures multi-scale temporal dynamics, feature-graph relations and regime shift at once is answered by the tri-stream design: the Multi-Scale TCN encodes local temporal structure at up to three resolutions, the Feature Graph Encoder propagates messages between feature nodes over a Pearson adjacency, the Regime Gating Embedding supplies the distributional view, and the SE-bottleneck Gated Fusion joins the three. Secondly, no baseline explicitly handles the MCO structural break is answered by pairing reversible instance normalisation, which removes the pre-window level and scale shift that a fixed training-set normaliser cannot, with a soft mixture over three learned regime embeddings whose gate is inferred from the input statistics alone, so no date flag or external MCO label is required at inference. Thirdly, all baselines treat the feature vector as flat is answered by Feature Group Fusion, which routes the target, lag, temporal and external index segments through separate encoders and weights them with a learned softmax gate, so structurally disparate sources do not enter the network with identical initial weight. The contribution of each module is quantified in the ablation study report in Chapter 4.

3.5.1 Feature Selection

All 79 inputs are organised into five semantic groups per Table 3.4 (targets, temporal, external, lag, static). Non-contiguous groups are defined as index segments. Each group is independently embedded, fused via a learned softmax gate, and projected to d_model before the three parallel encoders (see Figure 3.18).

<u>How Features are Used</u>

Step 1 – Semantic Group Embedding: Each group is independently projected by a small to a common per-group embedding dimension d/2:

$$Group\ g\colon\ x_g\ (B,\ T_{in},\ F_g)\ \rightarrow\ Linear(F_g,\ d/2)\ \rightarrow\ GELU\ \rightarrow\ (B,\ T_{in},\ d/2)$$

Independent embedding allows each group's internal structure to be learned without cross-group interference at the first projection stage. This is architecturally analogous to how a domain expert analyses each data source on its own before synthesizing insights across sources.

Step 2 – Gated group fusion with learned softmax weights: All group embeddings are concatenated and passed through a learned softmax group gate, which adaptively weights each group's contribution at each timestep:

$$h_{concat} = cat[h_1, \dots, h_5] \rightarrow (B, T_{in}, n_{groups} \times d/2)$$

$$gate = softmax\left(Linear(d_{concat} \rightarrow n_{groups})\right) \rightarrow (B, T_{in}, n_{groups})$$

$$h_{gated} = \Sigma_g(gate_g \cdot h_g) \rightarrow (B, T_{in}, n_{groups} \times d/2)$$

The softmax gate forces the model to explicitly allocate attention budget across feature groups at each timestep. On holiday dates, the temporal/cyclical group receives higher weight; on normal weekdays, the target context (recent ridership) and lag groups dominate. The gate learns this allocation from data without manual tuning.

Step 3 – Projection to $\boldsymbol{d_{model}}$: The gated fusion output is projected to d_{model} via:

$$Linear(d_{concat} \rightarrow d_{model}) \rightarrow GELU \rightarrow Dropout \rightarrow Linear(d_{model} \rightarrow d_{model}) \rightarrow LayerNorm$$

This two-stage projection with LayerNorm compresses the fused group embeddings into a unified representation of dimension d_{model} on which all three parallel encoders then operate.

3.5.2 Key Architecture Components

3.5.2.1 Reversible Instance Normalisation (RevIN)

RevIN is applied to the raw input before feature group embedding (Kim et al., 2023). Each sample's mean and variance are computed per feature and used to normalise the input:

$$\hat{x} = \left(x - \mu_{sample}\right)/\sigma_{sample} \times \gamma + \beta$$

where γ and β are learnable per-feature affine parameters. The primary motivation is to reduce within-batch distribution shift caused by the MCO regime change. Without instance-level normalisation, a single minibatch containing samples form multiple regimes causes gradient conflicts that impair convergence. RevIN denormalises predictions after the forecast heads and before loss computation, ensuring the loss is measured in the original MinMax-scaled space where the target scaler was fitted.

3.5.2.2 Multi-scale Temporal Convolution Network (TCN) – Temporal Encoder

The Multi-scale TCN applies casual dilated convolutions at up to three temporal scales:

- **Scale 1 (full T_{in}):** CausalConv blocks with exponentially increasing dilation capture short-range and medium-range temporal dependencies.
- **Scale 2 ($T_{in}//2$, activated for $T_{in} \geq 28$):** CausalConv on the first half of the window; captures medium-range structures.
- **Scale 3 ($T_{in}//4$, activated for $T_{in} \geq 56$):** CausalConv on the first quarter; capture long-range structural baselines.

WaveNet-style gated activations ($\tanh \odot \sigma$) regulate information flow within each causal block. DropPath (stochastic depth) stochastically drops entire residual paths during training, with linearly increasing rates from 0 to –drop-path argument. A learned attention pooling mechanism blends outputs across active scales, producing the final temporal summary $h_t \in \mathbb{R}^{d_{model}}$.

3.5.2.3 Spatial Graph Encoder

Rather than using the raw temporal sequence as node features (as graph-based baselines do), the Feature Graph Encoder first compresses the time dimension via learned temporal attention:

$$\hat{x}(B, T_{in}, F) \rightarrow \text{learned time-attn} \rightarrow (B, F, 1)\text{node feature vectors}$$

A 2-layer GCN propagates signals across the Pearson-correlation adjacency (|corr| ≥ 0.1, same construction as STGCN/ASTGCN), with global mean pooling over all F feature nodes producing h_{graph}. The graph branch receives a temporal summary rather than the raw sequence, decoupling cross-feature relational encoding from temporal dynamics. Graph and regime branches receive raw RevIN-normalised input, bypassing feature-group fusion (Figure 3.18).

3.5.2.4 Regime Gating Embedding

The MCO period represents a structural break in Malaysia transit ridership – not merely a large shock, but a change in which features predict ridership at all. During lockdown, fuel prices and population density features become irrelevant to transit demand (which was prohibited regardless). The Regime Gating Embedding maintains K=3 learnable embedding vectors representing the three operational regimes:

- **Pre-COVID:** 2019-01-01 to 2020-03-17
- **MCO:** 2020-03-18 to 2021-12-31
- **Post-MCO:** 2022-01-01 to 2025-12-31

At each forward pass, the mean-pooled input is projected to K=3 soft logits via a linear layer and softmax, producing a soft mixture of regime embeddings:

$$h_r = \Sigma_k gate_k \cdot E_k, \quad \text{gate} = \text{softmax}\left(\text{Linear}\big(\hat{x}.\text{mean}(T)\big)\right)$$

This soft gating allows the model to blend adjacent regime characteristics for transitional periods (e.g., early MCO easing in 2021) without hard chronological boundaries. Crucially, no date metadata is required at inference, the regime gate is inferred from the observed feature pattern alone.

3.5.2.5 Gated Fusion

The three encoder outputs h_t (temporal TCN), h_s (spatial TCN), and h_r (regime gating) are concatenated and passed through a Squeeze-and-Excitation (SE) bottleneck gate:

$$h = cat[h_t, h_s, h_r] \quad (B, 3 \cdot d_{model})$$

$$g = \sigma\big(\text{bottleneck}(h)\big) \quad SE: 3d \to 3d//4 \to 3d, \text{then sigmoid}$$

$$out = \text{Linear}(g \odot h \to d_{model}) \to \text{GELU} \to \text{Dropout} \to \text{LayerNorm}$$

The SE bottleneck avoids a computationally expensive 3d x 3d weight matrix while still learning to calibrate which of the tree encoder streams contributes most for each input context.

3.5.2.6 Dual Forecast Heads and Custom Loss

Two forecast heads are applied to the fused embedding:

- **Primary head:** $h \rightarrow \text{Linear}(d_{model}, d_{model}) \rightarrow \text{GELU} \rightarrow \text{Linear}(d_{model}, T_{out})$ with highway residual and LayerNorm.
- **Boosting head:** $h \rightarrow \text{Linear}(d_{model}, T_{out}) \times \text{sigmoid}(\alpha)$ where α is initialised to -2.0 so sigmoid(-2.0) ~ 0.12 suppresses the boosting contribution early in training; the head gradually activates as training converges

Final prediction: $y = y_{primary} + y_{boost}$

The custom loss penalises both prediction accuracy and smoothness:

$$L = \text{WeightedHuber}(\text{step-decay} \quad \gamma = 0.9) + \lambda \cdot \text{TemporalSmoothness}$$

Step 1 (1-day-ahead) has weight 1.0

Step 7 (7-day-ahead) receives weight 0.9^6 ~ 0.53.

TemporalSmoothness = $|y_{t+1} - y_t|^2$ across T_{out} prevents oscillatory day-to-day predictions.

3.5.2.7 Future Temporal Projection – Known-Future Calendar Conditioning

The 16 temporal features are deterministic for any future calendar date, so it was pre-computed for the T_{out} forecast steps of each window as X_{future}: $(B, T_{out}, 16)$. A per-step

MLP ($n_t \rightarrow \max(2 \cdot n_t, 32) \rightarrow 1$, zero-initialised so it starts as a no-op) learns an additive correction from this known-future context, letting the model anticipate weekend dips and holiday effects inside the forecast horizon rather than extrapolating them from the look-back window.

3.5.3 Architecture Diagram

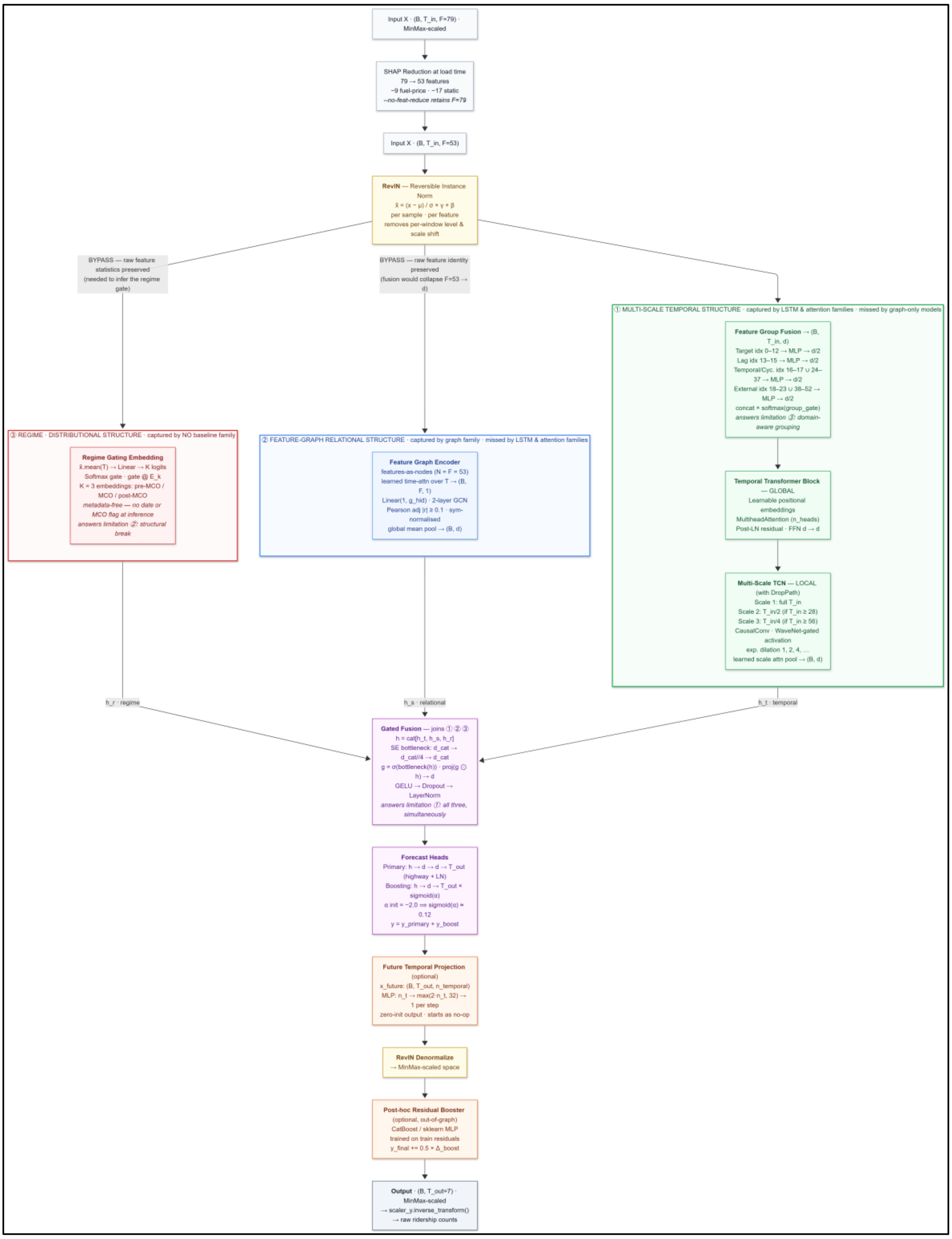

Figure 3.18 - HMT-TSF Architecture Organised by the Three Types of Structure the Model is Designed to Capture Simultaneously.

Note. Colour bands correspond to the three limitations listed at the start of Chapter 3.5. Band 1 is multi-scale temporal, captured by the LSTM and attention baseline families but not by graph-only models; band 2 is feature-graph relational, captured by the graph family but by neither the LSTM nor the attention family; band 3 is regime-distributional, captured by none of the fourteen baselines. The two BYPASS edges carry the instance-normalised input directly to the graph and regime streams: Feature Group Fusion projects the 53 feature channels into a 128-dimensional latent space, which would destroy the per-feature identity that a features-as-nodes graph requires and the per-feature statistics from which the regime gate is inferred. F = 53 shown for the feature-reduced variant (HMT-TSF-FR); the full variant retains F = 79 and skips the SHAP reduction block. It groups the architecture by the three types of structure rather than by layer order, so that each of the three limitations identified above maps to a visible region of the network: the green band to multi-scale temporal dynamics, the blue band to feature-graph relational structure, and the red band to regime-distributional shift.

3.5.4 Design Trade-offs

HMT-TSF jointly addresses temporal dynamics (multi-scale TCN), inter-feature correlations (GCN on Pearson adjacency), and MCO regime shift (RevIN + soft regime gating), with semantic group fusion allocating source-level weight per context. Key limitations remain: **(1)** higher parameter count and tuning complexity than any single baseline; **(2)** static Pearson graph sensitivity to distributional shift, partially offset by RevIN and regime gating; **(3)** manually specified K = 3 regime boundaries tied to

Malaysian MCO chronology; **(4)** optional post-hoc CatBoost residual correction adds a less interpretable second stage. Full training configurations are in Appendix C4.

3.6 Evaluation Metrics

3.6.1 Combined Accuracy Percentage (Combined %)

Combined% serves as a composite index that aggregates absolute, proportional, and squared error metrics to provide a single performance rank for model comparison. The composite value is computed as:

$$\text{Combined\%} = \max(0, 100 - \text{MAPE} - \text{MAE\%} - \text{RMSE\%})$$

- Higher is better, clipped, cannot be negative.
- All three constituent terms are mean-demand-normalized percentages, making them dimensionally homogeneous and directly addable on the same scale.

Equal weighting reflects the operational reality of transit management, where percentage errors relative to daily volume (MAPE), average absolute passenger discrepancies (MAE%), and large individual forecast failures (RMSE%) are equally critical.

3.6.2 Constituent Metrics

MAPE%, MAE%, RMSE%, and R^2 follow the definitions in Table 2.5. In this study, all percentage errors are mean-demand-normalised so MAPE%, MAE%, and RMSE% are dimensionally homogeneous and directly composable in Combined%. Raw MAE and RMSE (absolute passenger counts) are reported separately as operational diagnostics. R^2 measures explained variance and discriminates models with similar error magnitudes but different directional tracking. Together with Combined%, these five metrics evaluate HMT-TSF against the 14 baselines across look-back horizons and MCO conditions.

$$\text{MAPE} = \frac{1}{n}\sum_{i=1}^{n}\frac{|\hat{y}_i - y_i|}{|y_i|} \times 100$$

- y_i = actual ridership, $\hat{y}_i$ = predicted ridership
- Unit: %, used directly in Combined%
- Lower is better.

$$\text{MAE\%} = \frac{\sum_{i=1}^{n}|\hat{y}_i - y_i|}{\sum_{i=1}^{n} y_i} \times 100 \;=\; \frac{\text{MAE}}{\bar{y}} \times 100$$

- $\bar{y}$ = mean actual ridership across all test observations
- Unit: % - scale-free; used directly in Combined%
- Raw MAE (absolute ridership counts) is report separately as an operational diagnostic.
- Lower is better.

$$\text{RMSE\%} = \frac{\text{RMSE}}{\bar{y}} \times 100 \qquad \text{where} \qquad \text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(\hat{y}_i - y_i)^2}$$

- Same mean-demand denominator $\bar{y}$ as MAE%
- Unit: % - used directly in Combined%
- Raw RMSE (ridership counts) is reported separately as an operational diagnostic.
- Lower is better.

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2}$$

- y_i = actual ridership, $\hat{y}_i$ = predicted ridership, $\bar{y}$ = mean of actual values
- Numerator = Residual Sum of Squares (SSR) – unexplained variance
- Denominator = Total Sum of Squares (SST) – total variance in the data
- Range: -∞ to 1; $R^2 < 0$, indicates performance worse than predicting the mean.
- Higher is better.

3.7 Conclusion

This chapter established the complete methodological pipeline for forecasting daily Malaysian public transit ridership across twelve service lines and aggregate network demand. Eight heterogeneous data sources were collected, cleaned, and aligned into a unified 79-feature daily matrix under both MCO-inclusive (2019–2025) and MCO-exclusive (2022–2025) configurations. Feature engineering transformed this matrix into chronologically partitioned sliding-window tensors with training-only MinMax scaling, preserving temporal integrity and preventing information leakage.

Explanatory analysis confirmed that Malaysian ridership is shaped by distinct and sometimes non-linear forces: an 85–90% MCO collapse with mode-unequal recovery, spatially heterogeneous monsoon rainfall, heavy-rain modal substitution into transit, fuel-price-driven demand elasticity, and anticipatory holiday dips followed by return surges. Autoregressive lags at 7, 14, and 28 days dominate linear correlation structure, while static network and demographic features require non-linear modelling — motivating the deep-learning approach adopted herein.

14 baseline models were selected in a progressive comparison chain spanning spatiotemporal, graph-based, and attention-based architectures, each isolating a specific architectural contribution from sequential encoding through graph propagation to series decomposition. The proposed HMT-TSF forecaster addresses three gaps left by the baselines: simultaneous multi-scale temporal, graph-relational, and regime-aware encoding; explicit handling of the MCO distributional shift via RevIN and soft regime gating; and semantic feature-group fusion over the eight data sources. Model performance is evaluated using a composite Combined Accuracy index alongside MAPE%, MAE%,

RMSE%, and R^2, with all percentage errors mean-demand-normalised for direct comparability.

With the dataset, feature pipeline, model suite, and evaluation framework fully specified, Chapter 4 presents the experimental results — comparing all 15 models across look-back horizons and MCO conditions and assessing whether HMT-TSF achieves the forecasting accuracy required for proactive transit operations.

CHAPTER 4

FINDINGS & RESULTS

4.1 Experimental Evaluation

The central question is whether a purpose-built hybrid architecture can outperform established DL paradigms for 7-day Malaysia transit ridership forecasting, and under what conditions each paradigm succeeds or fails. First, default-configuration performance under normal operating conditions (MCO-excluded, 14-day look-back) establishes the baseline ranking. Second, hyperparameter tuning tests whether capacity increases translate into genuine generalisation gains. Third, look-back window sensitivity across 14, 28, and 56 days examines how much historical context a 7-day horizon requires. Fourth, structural-break robustness under the MCO pandemic period (2020-03-18 to 2021-12-31) stress-tests each model against an 85 – 90% ridership collapse. A dedicated fit-diagnostics analysis assesses generalisation quality independently of test-set accuracy.

All models follow the shared training protocol in Chapter 3 (AdamW, Huber loss, 70/15/15 chronological split). Performance is ranked by Combined%, where percentage terms are normalised by mean ridership demand; supplementary metrics (R^2, absolute MAE, RMSE) complete the profile. Headline configuration is base_nomco_lb14. Fit diagnostics use validation drift >25% above the best checkpoint and a train/validation gap ratio >3x.

4.2 Baseline Model Performance – Default Configuration (base_nomco_lb14)

Under the project default, baseline accuracy is tightly clustered: the top eight models occupy a 1.5 Combined% band, indicating a saturated regime in which architectural differences yield incremental rather than step-change gains. Table 4.1 presents the complete ranking with subsections below interpreting according to architectural series.

Table 4.1

Default-configuration (base_nomco_lb14) Performance of All 16 Baseline Variants, Ranked by Combined%

Rank	Model	Combined%	MAPE%	MAE%	RMSE%	R^2	MAE	RMSE
1	Informer	79.13	6.59	5.53	8.76	0.772	69,476	110,050
2	BiLSTM	79.04	6.52	5.77	8.67	0.776	72,460	109,007
3	TPA-LSTM	78.67	6.62	6.04	8.67	0.776	75,871	109,009
4	CNN-BiLSTM	78.18	6.81	6.12	8.89	0.764	76,884	111,782
5	LSTM	78.13	6.71	6.18	8.98	0.760	77,719	112,801
6	ST-LSTM	78.01	6.93	6.27	8.79	0.770	78,779	110,453
7	MTGNN	77.93	6.91	6.47	8.69	0.775	81,292	109,185
8	CNN-LSTM	77.64	6.92	6.32	9.12	0.752	79,375	114,632
9	STSGCN	76.97	7.18	6.42	9.42	0.736	80,703	118,391
10	STGCN	76.81	7.24	6.77	9.17	0.750	85,077	115,297
11	ASTGCN	75.69	7.42	6.89	9.99	0.703	86,631	125,606

12	CNN-LSTM-Augmented	75.36	7.77	7.19	9.68	0.721	90,356	121,639
13	Autoformer	74.73	7.86	7.51	9.90	0.708	94,414	124,408
14	CNN-LSTM-Parallel	74.18	8.18	7.68	9.97	0.704	96,477	125,240
15	STFGNN	73.94	8.21	7.60	10.25	0.688	95,547	128,777
16	PDR-STGCN	73.91	8.15	7.70	10.24	0.688	96,801	128,694

4.2.1 LSTM Models

Recurrent architectures are the strongest single-paradigm family for Malaysian ridership under normal conditions. Gated sequential memory aligns with the daily and weekly periodicity that Chapter 3.3's EDA identified as dominant demand drivers. LSTM-family models occupy five of the top eight positions, with BiLSTM (79.04%, $R^2 = 0.776$), TPA-LSTM (78.67%, $R^2 = 0.776$), and base LSTM (78.13%, $R^2 = 0.760$) establishing a strong performance floor for the comparison chain (Barath et al., 2026).

Bidirectional encoding provides a measurable advantage over unidirectional recurrence: BiLSTM's 0.91 Combined% gain over LSTM is consistent with bidirectional windows improving traffic congestion prediction by contextualising mid-sequence events (Krishnasamy et al., 2025). TPA-LSTM's near-equal R^2 to BiLSTM (0.776) validates temporal pattern attention over LSTM hidden states for subway passenger flow (Wei et al., 2023). CNN-BiLSTM (78.18%) and CNN-LSTM sequential model (77.64%) approach this band through hierarchical local-then-global encoding (Chen et al., 2022), while ST-LSTM (78.01%, $R^2 = 0.770$) ranks sixth with decoupled spatial encoding (Cui et al., 2025).

4.2.2 Graph Models

Graph-based models exhibit wider performance dispersion than LSTM models, reflecting heterogeneous strategies for encoding inter-feature correlations. MTGNN (77.93%, $R^2 = 0.775$) ranks seventh overall and leads the graph series, narrowly ahead of STSGCN (76.97%) and STGCN (76.81%). End-to-end learned asymmetric adjacency with multi-scale delated inception outperforms static correlation graphs for traffic flow prediction (Wu et al., 2025); MTGNN's placement is consistent with that finding, because asymmetric learned edges can represent directed relationships. For example, fuel-price signals influencing ridership without the reverse that symmetric Pearson graphs cannot.

STGCN benefits from Pearson-correlation feature-graph construction over the 79-feature input. Chebyshev graph convolution interleaved with temporal gated convolution captures relational dependencies missed by purely recurrent models (Deng, 2025). STSGCN (76.97%) slightly exceeds STGCN (76.81%), consistent with synchronous spatiotemporal graph operations capturing coordinated multi-feature regime shifts during public holidays (Chen et al., 2023). Lower-ranked graph variants (PDR-STGCN, STFGNN) are analysed under longer lookbacks (Chapter 4.4) and MCO stress (Chapter 4.5).

4.2.3 Attention Models

Informer (79.13%, R^2 = 0.772) is the single strongest baseline, narrowly ahead of BiLSTM (79.04%) by 0.09 Combined%. ProbSparse attention with a generative decoder matches recurrent accuracy while regularising against noise; at $T_{in} = 14$ the approximation is near-full attention (Chapter 3), so no accuracy penalty is expected from sparsity at this lookback. Informer's head-of-field placement, combined with superior MCO robustness (Chapter 4.5) and perfect base-configuration fit record (Chapter 4.6), makes it the most well-rounded baseline.

Autoformer (74.73%, R^2 = 0.708) ranks 13th, trailing leaders by a substantial margin. Stable Autoformer performance is reported on strongly periodic signals under appropriate sequence lengths (Ma & Zhang, 2025); here, MCO fragility (Chapter 4.5) further reduces practical utility. ASTGCN (75.69%, R^2 0.703) performs moderately, with its largest gains arriving only after tuning (+3.13 Combined%, Chapter 4.3) (Cui et al., 2025).

4.3 Effect of Hyperparameter Tuning

Hyperparameter tuning does not uniformly improve baseline performance. Instead, it produces model-specific accuracy shifts at a consistent generalisation cost documented in Chapter 4.6. Table 4.2 summarises tuned versus base results at nomco_lb14.

Table 4.2

Tuning Effect at tuned_nomco_lb14 Versus base_nomco_lb14

Model	Base Combined%	Tuned Combined%	ΔCombined
Informer	79.13	79.99	+0.86
TPA-LSTM	78.67	79.95	+1.28
BiLSTM	79.04	79.44	+0.40
ST-LSTM	78.01	79.13	+1.12
ASTGCN	75.69	78.82	+3.13
LSTM	78.13	78.43	+0.30
STGCN	76.81	78.39	+1.58
CNN-LSTM-Augmented	75.36	77.10	+1.74
PDR-STGCN	73.91	75.56	+1.65
Autoformer	74.73	75.21	+0.48
MTGNN	77.93	76.40	−1.54
CNN-BiLSTM	78.18	76.12	−2.06
STSGCN	76.97	73.65	−3.33
CNN-LSTM	77.64	73.36	−4.28
CNN-LSTM-Parallel	74.18	72.53	−1.65
STFGNN	73.94	65.47	−8.47

The median tuning improvement is approximately +0.5 Combined%, indicating that test-set accuracy is broadly flat while generalisation deteriorates. The largest beneficiaries are ASTGCN (+3.13, to 78.82%), CNN-LSTM-Augmented (+1.74), PDR-STGCN (+1.65), and STGCN (+1.58). ASTGCN and STGCN gains align with deeper attention and wider spatiotemporal graph configurations when regularisation is co-tuned (Cui et al., 2025; Deng, 2025).

The most severe regressions are STFGNN (-8.47, to 65.47%), CNN-LSTM sequential (-4.28), and STSGCN (-3.33). STFGNN's collapse – the study's largest tuning failure – indicates dual-graph fusion at this training budget memorises non-transferable patterns; CNN-LSTM sequential regresses as widened filters amplify an existing single-stream bottleneck. MTGNN (-1.54) and CNN-BiLSTM (-2.06) also regress, confirming that additional capacity harms some architecture. The best tuned baseline remains Informer (79.99%), still 5.79% Combined% below HMT-TSF – a gap Chapter 4.7 examines in detail.

4.4 Lookback Window Sensitivity

A systematic sweep across lb14, lb28, and lb56 reveals that longer historical context does not improve seven-day-ahead accuracy for most architectures. The principal finding is operational: lb14 is optimal for 10 of 16 models, lb28 for 6, and lb56 for none. Table 4.3 reports Combined% across the three windows under base_nomco.

Table 4.3

Look-back Sensitivity (Combined%) Across lb14, lb28, and lb56 for All Baseline Variants (base_nomco)

Model	lb14	lb28	lb56	Best

Informer	**79.13**	78.63	77.17	lb14
BiLSTM	**79.04**	76.97	77.09	lb14
TPA-LSTM	78.67	**79.08**	76.36	lb28
CNN-BiLSTM	**78.18**	72.65	74.72	lb14
LSTM	**78.13**	77.48	75.98	lb14
ST-LSTM	78.01	**78.86**	75.60	lb28
MTGNN	77.93	**78.11**	77.76	lb28
CNN-LSTM	**77.64**	73.90	74.33	lb14
STSGCN	**76.97**	75.02	75.15	lb14
STGCN	76.81	**77.68**	74.93	lb28
ASTGCN	75.69	**76.89**	75.15	lb28
CNN-LSTM-Augmented	**75.36**	74.73	74.04	lb14
Autoformer	**74.73**	73.61	73.40	lb14
CNN-LSTM-Parallel	74.18	**76.55**	74.33	lb28
STFGNN	**73.94**	61.30	43.25	lb14
PDR-STGCN	**73.91**	72.35	72.20	lb14

Note. Best window per model is shown in bold.

MTGNN is the most window-stable model (77.93 / 78.11 / 77.76), consistent with existing research where multi-scale dilated inception preventing any single temporal scale from dominating (Wu et al., 2025). STFGNN exhibits the most severe collapse: Combined% falls from 73.94% at lb14 to 43.25% at lb56, with R^2 of −0.112 — worse than naïve mean prediction — because the temporal adjacency matrix constructed from 56-timestep profiles supplies misleading structural priors (Chang et al., 2025). HMT-TSF peaks at lb14 under both MCO conditions (Chapter 4.7), reinforcing the 14-day recommendation for seven-day horizons.

4.5 MCO Structural Break Robustness

The MCO period represents a severe distributional discontinuity in which daily boardings feel from 800,000 – 1,200,000 to 50,000 – 200,000 under mobility restrictions. Including this window (mco configuration) tests whether models trained on pre-break, break, and post-break data can forecast through the transition. The degradation metric ΔCombined% (nomco → mco) serves as the primary robustness indicator; Table 4.4 ranks baselines at lb14.

Table 4.4

MCO Robustness at lb14 for All 16 Baseline Variants, Ranked by Combined% (Least to Most Degradation)

Model	nomco	mco	ΔCombined%	mco R^2
Informer	79.13	73.25	−5.88	0.705
BiLSTM	79.04	72.74	−6.29	0.699
CNN-LSTM-Parallel	74.18	66.89	−7.29	0.577
TPA-LSTM	78.67	71.05	−7.61	0.669
STGCN	76.81	68.55	−8.27	0.615
MTGNN	77.93	67.33	−10.60	0.581
PDR-STGCN	73.91	63.10	−10.81	0.494
CNN-LSTM-Augmented	75.36	64.50	−10.86	0.531
ST-LSTM	78.01	66.75	−11.26	0.584
STSGCN	76.97	64.20	−12.77	0.531
ASTGCN	75.69	61.69	−14.00	0.471
CNN-BiLSTM	78.18	60.59	−17.59	0.427
LSTM	78.13	58.92	−19.21	0.390
Autoformer	74.73	51.26	−23.47	0.121

CNN-LSTM	77.64	45.58	−32.06	−0.050
STFGNN	73.94	41.00	−32.94	−0.306
HMT-TSF	**85.78**	**81.99**	**−3.79**	**0.859**

Note. HMT-TSF included for reference.

The degradation spread among baselines runs from -5.88 (Informer) to -32.94 (STFGNN). Informer and BiLSTM are the most robust recurrent baselines; Informer's sparse attention to reduced sensitivity to distributional anomalies because only the most predictive query-key pairs are engaged across regime boundaries (Song et al., 2024). Two models fail in a technically meaningful sense: CNN-LSTM sequential (R^2 = -0.050) and STFGNN (R^2 = -0.306) produce forecasts less accurate than predicting the training-set mean – despite CNN-LSTM ranking eighth under normal conditions. Sequential CNN-LSTM can memorise local temporal patterns at the expense of adaptability, consistent with the -32.06 Combined% degradation observed here (Topilin et al., 2025).

STGCN demonstrates unexpectedly strong MCO robustness (Δ-8.27, mco R^2 = 0.615), outperforming more expressive models including MTGNN (Δ-10.66). STGCN's fixed Chebyshev graph constrains prediction drift when input distributions shift – a rigidity that limits peak accuracy but caps catastrophic failure. The broader expressiveness-stability trade-off is developed in Chapter 5.1.4; HMT-TSF addresses it through RevIN and regime gating (Chapter 4.7).

4.6 Generalisation and Overfitting Analysis

Test-set Combined% alone is an unreliable deployment criterion because several high-ranking baselines achieve accuracy partly through memorisation. Fit diagnostics across all 12 configurations per model (base and tuned × nomco and mco × three lookbacks) show that all 6 LSTM models and both simple-convolution graph models (STGCN, PDR-

STGCN) record zero good-fit verdicts, 0/12 each, despite strong headline scores for BiLSTM, TPA-LSTM, and CNN-BiLSTM.

The gap-ratio, best validation loss divided by minimum training loss, is the clearest signal. LSTM gap ratios span 3.3x to 13.6x, with CNN-BiLSTM-base under MCO at lb56 reaching 13.51x. PDR-STGCN is the most extreme memoriser, posting 38.61x at tuned_nomco_lb56, indicating training loss near zero while validation loss remains elevated. By contrast, Informer achieves a perfect 6/6 good-fit rate at base configuration – the only baseline with a clean record – with gap ratios of 2.16x – 2.98x consistently below the 3x threshold.

Among graph models, MTGNN (3/6 good-fit at base, 4/6 at tuned) and STSGCN (4/6 base) are cleanest after Informer, consistent with intrinsic regularisation from end-to-end learned adjacency (Wu et al., 2025). HMT-TSF leads the full study at 19/20 good-fit verdicts across both full and feature-reduced variants (Chapter 4.7). Tuning uniformly harms diagnostics: the good-fit rate falls from 26% (base) to 14% (tuned) across all 14 baselines, confirming that expanded capacity raises gap ratios without equivalent accuracy gains.

4.7 HMT-TSF Model Performance

HMT-TSF was evaluated across ten configurations (two MCO conditions × five lookbacks: 7, 14, 28, 56, 84 days). Table 4.5 presents the full results.

Table 4.5

HMT-TSF (full, feature = 79) Results Across All Ten Configurations.

Configuration	Combined%	MAPE%	MAE%	RMSE%	R^2	MAE	RMSE
nomco_lb7	84.88	4.74	4.25	6.13	0.888	53,325	76,951
nomco_lb14	**85.78**	**4.39**	**3.97**	**5.87**	**0.897**	**49,897**	**73,757**
nomco_lb28	84.23	4.94	4.55	6.28	0.883	57,268	78,910
nomco_lb56	82.53	5.52	5.11	6.84	0.859	64,198	85,874
nomco_lb84	77.68	6.99	6.59	8.73	0.773	82,174	108,800
mco_lb7	78.61	6.89	6.12	8.38	0.807	75,401	103,170
mco_lb14	**81.99**	**5.68**	**5.18**	**7.15**	**0.859**	**63,853**	**88,228**
mco_lb28	79.89	6.26	5.83	8.02	0.822	72,032	99,059
mco_lb56	79.47	6.68	5.91	7.95	0.825	73,334	98,598
mco_lb84	78.45	6.73	6.17	8.64	0.791	76,593	107,293

Note. Bold rows denote best nomco and mco configurations.

The feature-reduced variant exceeds these values at every nomco lookback ≥ 14 (Chapter 4.9).

Performance peaks at lb14 under both MCO conditions (nomco 85.78%, mco 81.99%), declining at longer windows once X_{future} supplies explicit future calendar structure. At nomco_lb14, the full deployable system (with X_{future}) leads every baseline on every headline metric — +6.65 Combined% over Informer (79.13%) and 28.2% lower MAE (49,897 versus 69,476); the best tuned baseline (Informer, 79.99%) remains 5.79 points below HMT-TSF and 6.60 below HMT-TSF-FR (86.59%, Chapter 4.9). HMT-TSF alone receives the known-future calendar tensor (X_{future}); Chapter 4.7.1 quantifies the conditioning-versus-architecture split. These gains reflect the hybrid design in Section 3.5, which no single-paradigm baseline implements.

HMT-TSF with X_{future} is also the most MCO-robust model at lb14 (Δ -3.79 versus Informer -5.88), retaining the highest absolute MCO accuracy (81.99%) despite weaker FR robustness under structural break (Chapter 4.9). It posts 19/20 good_fit verdicts with X_{future} enabled (gap ratios 1.57x – 2.64x); only nomco_lb84 is overfit (+36.9% validation drift). Without X_{future}, each variant records 9/10 good-fit verdicts, with nomco_lb56 overfit for both full and FR models (Chapter 4.7.1).

Three sequential non-overlapping out-of-sample blocks test whether headline performance depends on a single favourable temporal partition. The evidence shows that no block falls below 71.99 Combined% or 0.707 R^2 across all ten with-X_{future} configurations; at nomco_lb14, block Combined% is 90.62, 82.14, and 85.12, with Block 2 weakest under nomco and Block 1 (lockdown-adjacent) weakest under mco. The analysis further shows that mco_lb84 is the most temporarily stable configuration (block spread 3.7 Combined%) while nomco_lb14 spreads 8.5 points, linking peak-accuracy deployment to nomco_lb14 or HMT-TSF-FR and cross-period consistency to longer MCO-trained windows at the accuracy cost in Table 4.5.

4.7.1 Known-Future Calendar Ablation (--no-x-future)

To isolate the contribution of known-future calendar conditioning, all 10 HMT-TSF configurations were retrained with –no-x-future. Baselines were not given equivalent X_{future} inputs, so this ablation measures HMT-TSF with and without its future-calendar projection module under an otherwise identical protocol.

Table 4.6

X_{future} Ablation at lb14. MCO Δ = Combined% Change from nomco_lb14 to mco_lb14 for The Same Variant and X_{future} Setting

Variant	X_future	nomco_lb14 Combined%	nomco_lb14 R^2	nomco_lb14 MAE	Δ vs Informer (79.13%)	mco_lb14 Combined%	MCO Δ (nomco→mco)
HMT-TSF (full)	On	85.78	0.897	49,897	+6.65	81.99	−3.79
HMT-TSF (full)	Off	80.89	0.790	62,836	+1.76	75.49	−5.40
HMT-TSF-FR	On	86.59	0.906	46,323	+7.46	78.74	−7.85
HMT-TSF-FR	Off	81.23	0.802	62,547	+2.10	74.01	−7.22
Informer (baseline)	—	79.13	0.772	69,476	—	73.25	−5.88

First, without X_{future}, HMT-TSF still leads Informer at nomco_lb14 (+1.76 Combined% full, +2.10 FR) and cuts MAE by approximately 10% (versus 28-33% with X_{future}) – a modest but positive architecture-only margin within the 1.5 Combined% baseline saturation band (Chapter 4.2). Second, X_{future} is not merely a deployment convenience: disabling it worses MCO lb14 degradation for the full model (-5.40 versus -3.79 with X_{future}), indicating that future-known holiday and weekend structure also aids structural-break transfer. Third, lb14 remains the optimum under both conditions, but nomco_lb56 becomes overfit for both full and FR ablation runs (replacing nomco_lb84 as the sole overfit case for the full model when X_{future} is on), linking calendar conditioning to generalisation at intermediate lookbacks.

Table 4.7

HMT-TSF (full, F=79) Without X_{future} Across All 10 Configurations

Configuration	Combined%	MAPE%	MAE%	RMSE%	R²	MAE	Fit verdict
nomco_lb7	78.47	6.03	5.52	7.98	0.759	73,940	good_fit
nomco_lb14	**80.89**	**5.72**	**5.00**	**8.39**	**0.790**	**62,836**	**good_fit**
nomco_lb28	80.87	5.73	5.03	8.37	0.794	63,216	good_fit
nomco_lb56	79.46	6.12	5.62	8.80	0.781	70,395	overfit
nomco_lb84	76.32	6.99	6.59	9.29	0.715	79,639	good_fit
mco_lb7	74.60	7.41	6.79	9.20	0.697	86,066	good_fit
mco_lb14	75.49	7.18	6.59	9.73	0.716	83,592	good_fit
mco_lb28	74.84	7.32	6.74	9.10	0.704	86,528	good_fit
mco_lb56	**76.02**	**7.01**	**6.44**	**9.53**	**0.723**	**80,440**	**good_fit**
mco_lb84	74.82	7.12	6.58	9.50	0.705	85,938	good_fit

4.8 SHAP Feature Importance Analysis

SHAP values were computed for the trained nomco_lb14 model using shap.GradientExplainer with 100 test samples as background. Absolute SHAP values were averaged over samples, time steps, and forecast horizons to produce a 79-element importance vector resolved from feature_metadata.json.

The highest-importance features align with established transit demand theory. Annual-cycle encodings dominate attribution magnitude: month_cos, month_sin, and month carry the three largest mean |SHAP| values, anchoring seven-day forecasts on seasonal position – school terms, festive seasons, and the northeast monsoon. total_ridership is the strongest non-seasonal driver and the only feature in the top 15 of all ten configurations (100% consistency), confirming target autocorrelation as the dominant data-driven signal. Holiday structure forms the second calendar tier: days_to_next_public_hol ranks in the headline top 10 and appears in 80% of configurations, quantifying the pre-holiday dips and post-holiday surges identified in Chapter 3.3's EDA. Among external covariates, unfrozen fuel grades (fp_lv_diesel, fp_lv_ron97) and monsoon-corridor rainfall (Perlis, Terengganu, Penang) complete the headline top 10 – consistent with multi-source fuel and weather signals as secondary urban rail demand drivers (Cui et al., 2025).

Cross-configuration aggregation reveals 23 zero-SHAP features plus three near-zero RON95 variants, defining the 26-feature removal set for HMT-TSF-FR (Chapter 4.9). Their persistent inactivity independently validates the EDA conclusion in Chapter 3.3 that administered RON95 series and time-invariant spatial descriptors carry no day-to-day discriminative variance in the post-MCO training window. Importance also shifts with lookback: calendar features dominate at short windows, while ridership autocorrelation rises at lb56 – lb84 – so a single SHAP run should not be treated as the definitive driver hierarchy without cross-configuration corroboration.

4.9 Feature-Reduced Variant: HMT-TSF-FR

Removing 26 zero- or near-zero-SHAP features, the entire static group plus 9 near-constant administered fuel columns, produces HMT-TSF-FR on a 53-feature input. Static descriptors are redundant because time-invariant structural capacity is implicitly encoded in historical ridership given sufficient look-back; administered fuel columns carry near-zero day-to-day variance across the post-MCO training window.

HMT-TSF-FR achieves Combined% of 86.59 at nomco_lb14 (R^2 = 0.906, MAE = 46,323), exceeding the full model (85.78) by +0.81 and setting the study-wide headline. The graph adjacency shrinks from 79 x 79 to 53 x 53 and explainability memory decreases by approximately 33%. The FR advantage grows with look-back (+0.81 at lb14, +1.84 at lb28, +1.52 at lb56, +4.26 at lb84), repairing nomco_lb84's overfit verdict under feature reduction.

The trade-off emerges under structural break: HMT-TSF-FR loses to the full model in every mco configuration (Δ -0.85 at lb7 to -5.22 at lb84). Near-constant fuel-price columns evidently stabilise predictions under lockdown-spanning distributional shift; the full model should be retained for shock-prone conditions while FR is recommended for normal operations. All HMT-TSF-FR configurations remain good_fit with gap ratios within 1.66x – 2.72x, confirming that feature reduction does not introduce overfitting.

4.10 Summary

First, the full HMT-TSF (with X_{future}) outperforms all 16 baselines at nomco_lb14 (Combined% 85.78; 86.59 for FR), leading Informer (79.13%) by 6.65 points and reducing MAE by 28.2% (33.3% for FR). Ablating X_{future} retains a modest lead over Informer within the saturated baseline band (Chapter 4.7.1).

Second, HMT-TSF with X_{future} is the most MCO-robust and most generalisable configuration. MCO degradation at lb14 is Δ -3.79 (versus -5.40 without X_{future} for the full model); retained MCO accuracy (81.99%) leads the field; it also posts the study's strongest good-fit record (Chapter 4.6-4.7).

Third, LSTM models dominate the top eight yet overfit systematically. Recording zero good-fit verdicts across all 12 configurations. Informer is the sole baseline with a clean base fit record and the most deployable alternative.

Fourth, lb14 is the decisively optimal for a 7-day horizon. Longer windows trigger severe collapses for fragile architectures (Chapter 4.4). SHAP analysis (Chapter 4.8) enables principled reduction to 53 features for normal-regime deployment, with the full 79-feature model retained for shock-prone regimes (Chapter 4.9).

CHAPTER 5

DISCUSSION & CONCLUSION

5.1 Discussion of Key Findings

5.1.1 Hybrid Multi-Modal Architectures Outperform Single-Paradigm Models

Principled fusion of complementary inductive biases outperforms any single architectural paradigm for 7-day Malaysian transit ridership forecasting when the full deployable system is evaluated. HMT-TSF achieved Combined% 85.78 and R^2 0.897 at nomco_lb14 (86.59 / 0.906 for FR), leading Informer (79.13%) by 6.65 Combined% and cutting daily MAE by 28.2%, with the best tuned baseline (Informer, 79.99%), still 5.79 Combined% behind. HMT-TSF alone receives known-future calendar conditioning (X_{future}), legitimately available at deployment but absent from baseline inputs; Chapter 4.7.1 quantifies how much of this margin reflects conditioning versus hybrid architecture alone. Within the 1.5 Combined% baseline saturation band (Chapter 4.2), marginal gains therefore lie in multi-modal fusion plus task-appropriate input design rather than incremental paradigm refinement, a finding that directly motivates the generalisation and robustness analyses that follow.

5.1.2 Test-set Accuracy Conceals Systematic Overfitting in Recurrent Baselines

Nominal test-set accuracy is an unreliable indicator of deployment readiness because the strongest recurrent baselines achieve their rankings partly through memorisation: Chapter 4.6 records zero good-fit verdicts for all 6 LSTM models and both simple-convolution graph models across 12 configurations (train/validation gaps 3.3x – 13.6x; PDR-STGCN 38.61x) despite BiLSTM and TPA-LSTM ranking second and third (R^2 0.776). This dissociation on memorisation at the expense of adaptability (Topilin et al., 2025) and contrasts Informer's unique 6/6 clean-fit record via ProbSparse regularisation (Song et al., 2024) – had only test Combined% been reported, BiLSTM would appear nearly equivalent to Informer, yet diagnostics place them at opposite ends of the generalisation spectrum – while hyperparameter tuning further eroded fit quality (good-fit rate 26% to 14%) for small, inconsistent test gains. For deployment, fit diagnostics must therefore be evaluated alongside accuracy.

5.1.3 14-Day Look-Back Window Is Decisively Optimal Under Normal Conditions

For a 7-day forecast horizon under normal operating conditions, a 14-day historical window is optimal and longer context actively harms most architectures: across Chapter 4.4, lb14 was best for 10 of 16 baseline variants (lb28 six, lb56 none), and HMT-TSF with X_{future} was single-peaked at lb14 (85.78%). Fragile architectures collapse at longer windows when structural priors mislead (Chang et al., 2025), whereas MTGNN's multi-scale dilated inception remains near-stable (Wu et al., 2025). Ridership predictability concentrates in two weekly cycles, with slower seasonal structure – dominant in SHAP attribution (Chapter 4.8) – supplied through calendar features rather than extended history (Ma & Zhang, 2025). HMT-TSF retains its lb14 optimum in both MCO and nomco regimes under both X_{future} settings which transitions naturally to structural-break robustness.

5.1.4 Structural Inductive Biases Determine Robustness Under Distributional Shift

Robustness to structural breaks is governed less by nominal accuracy than by the rigidity or adaptivity of a model's structural assumptions. Chapter 4.5 documents Δ from HMT-TSF's −3.79 (with X_{future}) to STFGNN's −32.94, with two baselines falling below naïve mean-prediction despite strong default-configuration rankings. Disabling X_{future} worsens full-model MCO degradation at lb14 to Δ −5.40, indicating that future-known holiday and weekend structure contributes to structural-break transfer, not only to nominal nomco accuracy. This reveals an expressiveness–stability trade-off: constrained designs (STGCN (Deng, 2025)) and sparse attention (Informer (Song et al., 2024)) transfer more reliably than flexible learners that memorise pre-break regularities. HMT-TSF with X_{future} resolved this through RevIN and regime gating embedding (81.99% MCO accuracy, clean fit under break), as analysed further in Chapter 4.9.

5.1.5 Learned Feature Hierarchy Is Domain-Consistent and Enables Principled Feature Reduction

The model's learned feature importance is interpretable, theoretically coherent, and actionable: SHAP analysis (Chapter 4.8) recovers a domain-consistent hierarchy, seasonal calendar structure, ridership autocorrelation, holiday anticipation, and secondary fuel/weather signals (Cui et al., 2025). The practical consequence is HMT-TSF-FR, removing 26 zero- or near-zero-SHAP features improved accuracy at every nomco lookback ≥14 (86.59 Combined%, 53x53 graph), as static spatial descriptors are redundant given time-invariant capacity encoded in historical ridership, though FR's consistent MCO losses (-0.85 to -5.22) reinforce that feature decisions validated under normal conditions must be re-validated under regime shift.

5.2 Implications of the Study

5.2.1 Theoretical Implications

First, it shows that the features-as-nodes graph formulation transfers graph-based traffic methods (Chang et al., 2025; Deng, 2025; Wu et al., 2025) to multivariate national ridership data, while exposing where transfer breaks down (temporal-adjacency construction at long windows). **Second**, it demonstrates that fit diagnostics constitute a necessary complement to test-set accuracy, materially reordering practical model rankings relative to accuracy-only evaluation. **Third**, it establishes regime gating combined with RevIN as an effective mechanism for structural-break robustness, extending regime-shift observations from attention sparsity (Song et al., 2024) into an explicit, learnable regime representation.

5.2.2 Practical Implications for Malaysian Transit

For established networks such as the Klang Valley and Penang systems, the findings support phased integration: HMT-TSF can run in parallel with existing planning processes, with forecast-informed decisions adopted incrementally as walk-forward validation (Chapter 4.7) accumulates confidence. For developing corridors in Johor and East Malaysia, where new services can be planned around demand forecasts from the outset, the system serves as a proof of concept for demand-responsive network design.

From an operational perspective, accurate 7-day demand forecasts enable frequency setting, rolling-stock allocation, and staffing to match predicted demand. Capacity-planning gains should be quoted against the deployment configuration analysed in Chapters 4.7 and 4.9 (full model with X_{future} for shock-prone regimes; HMT-TSF-FR for normal operations). These forecast errors give planners a quantitative basis for weekly capacity decisions (Hassan et al., 2025).

5.3 Limitations of the Study

First, this study operates at national daily granularity with service lines as features and does not address station-level or intra-day crowd management (Wei et al., 2023; Cui et al., 2025). **Second**, evaluation relies on one national dataset; although walk-forward validation supports temporal stability (Chapter 4.7), cross-network external validity remains untested. **Third**, inputs are batch-historical, so collection delays and source errors can be imputed but not prevented upstream. **Fourth**, the MCO period supplies only one observed structural break, leaving the K = 3 regime gating embedding calibrated to a single shock type. **Fifth**, all headline results are single-seed point estimates; without repeated-seed variance or formal significance tests, sub-percentage gaps within the 1.5 Combined% saturation band should be treated as indicative. **Sixth**, the study models demand-side unreliability only and excludes supply-side disruptions (delays, breakdowns, cancellations) for which no public Malaysian daily dataset exists. **Seventh**, the MCO-excluded setting still carries indirect MCO information through ridership_lag_{7,14,28} at the post-MCO sample start. **Eighth**, HMT-TSF alone receives per-horizon known-future calendar input (X_{future}); Chapter 4.7.1 quantifies the effect, but no baseline received equivalent tensors, so strict architecture comparison remains partially confounded.

5.4 Future Works

Statistical validation and symmetric input design. Future work should re-run headline configurations across multiple seeds with bootstrap or Diebold-Mariano tests, and either equip top baselines with the same per-horizon future-calendar tensors used by HMT-TSF or retrain all models under a shared no-x-future protocol. Both arms would separate input-design gains from architecture gains and test whether sub-percentage ranking gaps are statistically robust.

Supply-side reliability data integration. AVL traces, on-time-performance records, and disruption logs could link demand forecasting to service reliability management.

Real-time data integration. Service alerts, trip updates, and vehicle positions would move the system from offline planning toward operational decision support.

Data integrity at the source. Upstream validation and reconciliation across fare-collection channels would reduce imputation burden and strengthen the autocorrelation signals.

Spatial and temporal disaggregation. Extending the features-as-nodes design to station-level nodes with geographic adjacency would enable intra-day forecasting and align the framework with station-level passenger flow research.

Generalisation to new service lines and networks. Scalability should be tested on new lines and networks, including cold-start and transfer-learning warm starts.

Regime modelling beyond the MCO. An open-set regime detector – recognising fare reforms or new-line openings online – would turn retrospective robustness into prospective deployment capability.

5.5 Final Verdict

This study set out to determine whether a purpose-built hybrid architecture could outperform established DL paradigms for 7-day Malaysian transit ridership forecasting, and to characterise the conditions under which each paradigm succeeds or fails. The answer is affirmative and well-bounded: HMT-TSF (with X_{future}) leads all baselines on headline metrics and structural-break robustness (Chapters 4.7 – 4.9), while a modest architecture-only margin remains without X_{future} (Chapter 4.7.1). Deployment should follow the lb14 window, fit-diagnostic, and feature-reduction rules established in Chapters 4.4, 4.6, and 4.9. Together, these results establish a credible foundation for demand-responsive transit planning in Malaysia.

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APPENDICES

APPENDIX A1 – Hardware Specification

- CPU: 64 cores
- GPU: 1x A100 (UMHPC) – 32GB vRAM
- No hyperthreading

APPENDIX A2 – Data Configuration

- Look-back window (T_{in}): 14 / 28 / 56 days for the 14 base models; 7 / 14 / 28 / 56 / 84 days for HMT-TSF.
- Forecast horizon (T_{out}): 7 days
- Data split: 70% train / 15% test / 15% validation (chronological)
- Precision: Sequences stored as float16, cast to float32 before model input
- Scaling: MinMaxScaler fitted on training data only

APPENDIX B1 – Baseline Models Hyperparameters

- Epochs: 150
- Batch size: 32
- LR: 1e-3
- Patience: 15
- Dropout: 0.1
- Weight decay: 1e-4

Some per-model deviations differ from standardized hyperparameters for better comparison according to its architecture:

- STGCN (Dropout: 0.20, Weight decay = 2e-4)
- STSGCN (Epochs: 200, LR: 5e-4, Patience = 25)
- STFGNN (Dropout: 0.20, Weight decay = 3e-4)
- PDR-STGCN (Dropout: 0.15, Weight decay = 2e-4)
- ASTGCN (Dropout: 0.15, Weight decay = 2e-4)
- Autoformer (Dropout: 0.15, Weight decay = 2e-4)

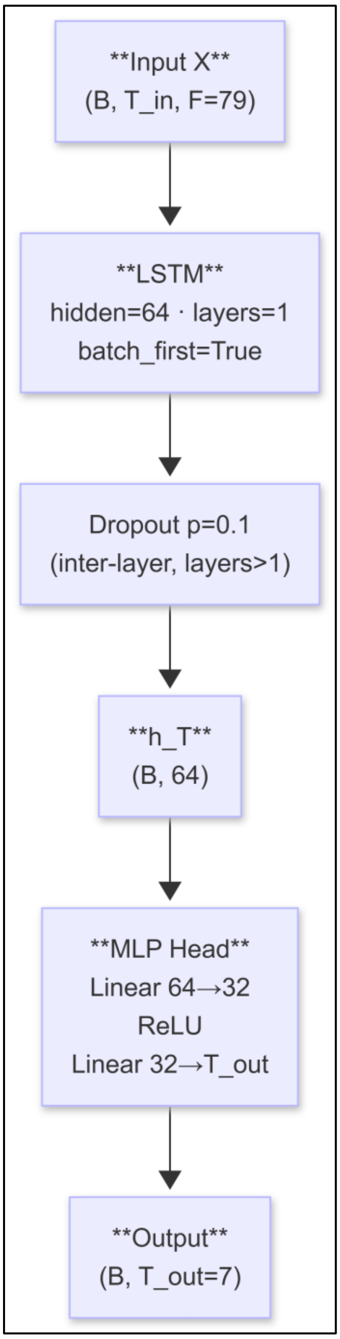
APPENDIX B2 – Baseline Models Training Optimization

- Optimiser: AdamW (decoupled weight decay)
- Loss: HuberLoss (delta = 1.0, default) – robust to ridership outliers
- LR schedule: Linear warmup → ReduceLROnPlateau

APPENDIX C1 – Spatiotemporal Models

LSTM – Long Short-Term Memory

A stacked LSTM reads the T_{in}-step look-back window left-to-right, compressing the entire input sequence into a single final hidden state h_T. This hidden state is passed directly to a two-layer MLP head to produce T_{out} simultaneous predictions. When layers > 1, dropout is applied between stacked LSTM layers.

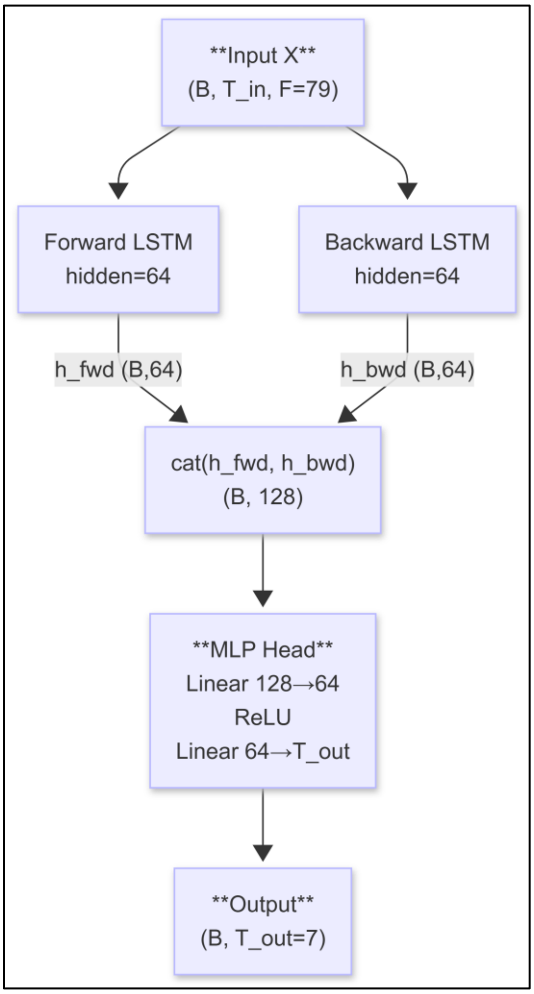


Hyperparameter changes for tuned variant:

- hidden $64 \rightarrow 128$
- layers $1 \rightarrow 2$
- dropout $0.10 \rightarrow 0.20$
- weight_decay $1e-4 \rightarrow 2e-4$
- Structure unchanged

BiLSTM – Bidirectional LSTM

Extends LSTM with a reversed recurrent pass over the same look-back window. The forward final hidden state h_{fwd} and backward final hidden state h_{bwd} (both from the last layer) are concatenated before the MLP head, doubling representational capacity.

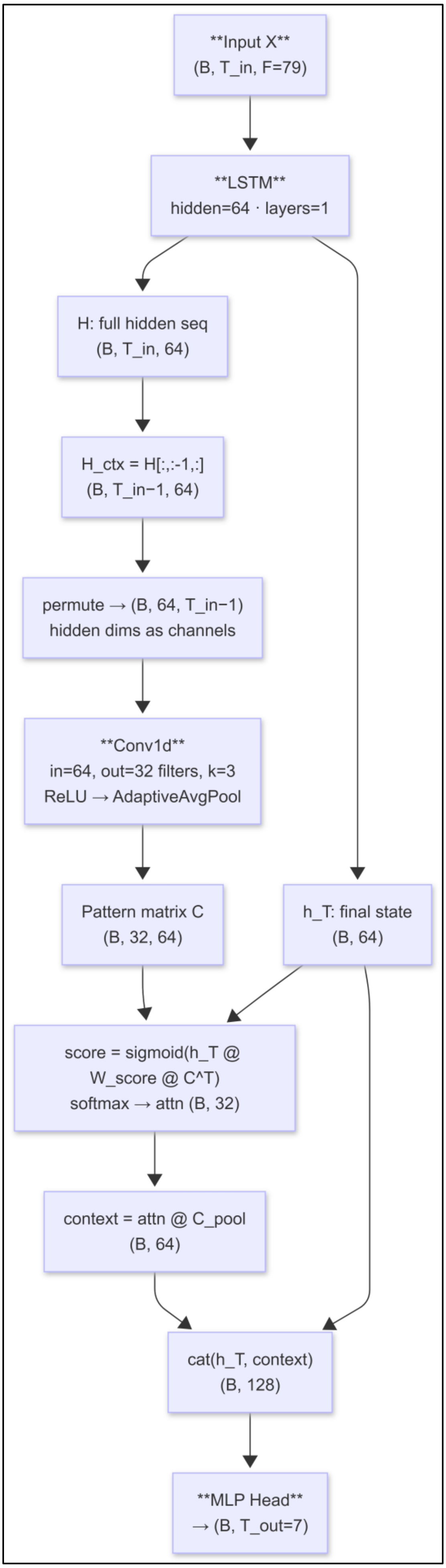


Hyperparameter changes for tuned variant:

- hidden 64 → 256
- layers 1 → 3
- dropout unchanged
- MLP head input dimension grows to 512 (bidirectional concatenation).
- Structure unchanged

TPA-LSTM – Temporal Pattern Attention LSTM

A standard LSTM produces both full hidden-state sequence H and the final hidden state h_T. The TPA module takes $H_{ctx} = H[:, : -1, :]$ (all but the last step), transposes it to treat hidden dimensions as channels over time, applies a 1-D CNN to extract temporal pattern filters, adaptive-average-pools across time, scores each filter against h_T via a learned scoring matrix W_{score}, and softmax-weights the scores to produce a pattern context vector. The concatenation of h_T and context is passed to the MLP head.



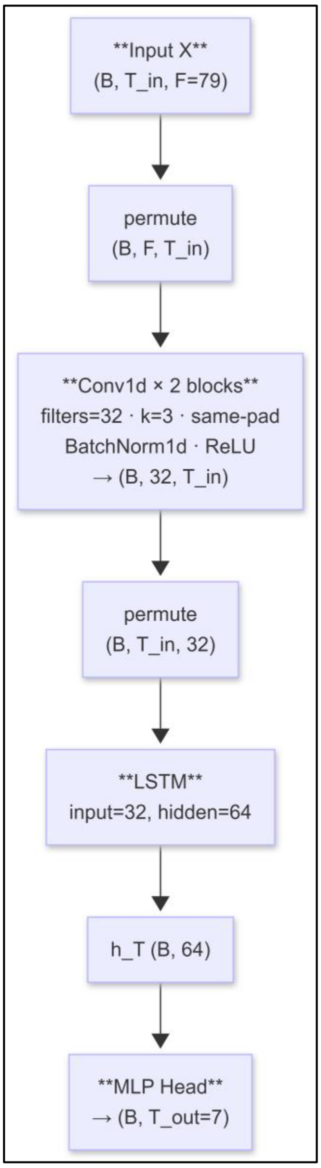
Input X
(B, T_in, F=79)
LSTM
hidden=64 · layers=1
H: full hidden seq
(B, T_in, 64)
H_ctx = H[:,:-1,:]
(B, T_in−1, 64)
permute → (B, 64, T_in−1)
hidden dims as channels
Conv1d
in=64, out=32 filters, k=3
ReLU → AdaptiveAvgPool
Pattern matrix C
(B, 32, 64)
h_T: final state
(B, 64)
score = sigmoid(h_T @
W_score @ C^T)
softmax → attn (B, 32)
context = attn @ C_pool
(B, 64)
cat(h_T, context)
(B, 128)
MLP Head
→ (B, T_out=7)

Hyperparameter changes for tuned variant:

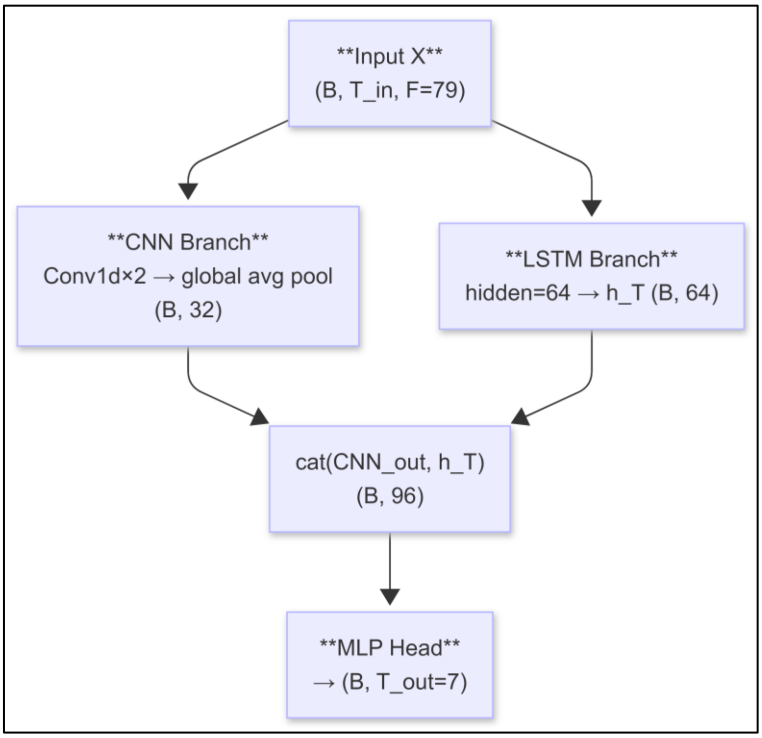
- hidden 64 → 256
- filters 32 → 128
- dropout 0.10 → 0.15
- Structure unchanged

CNN-LSTM

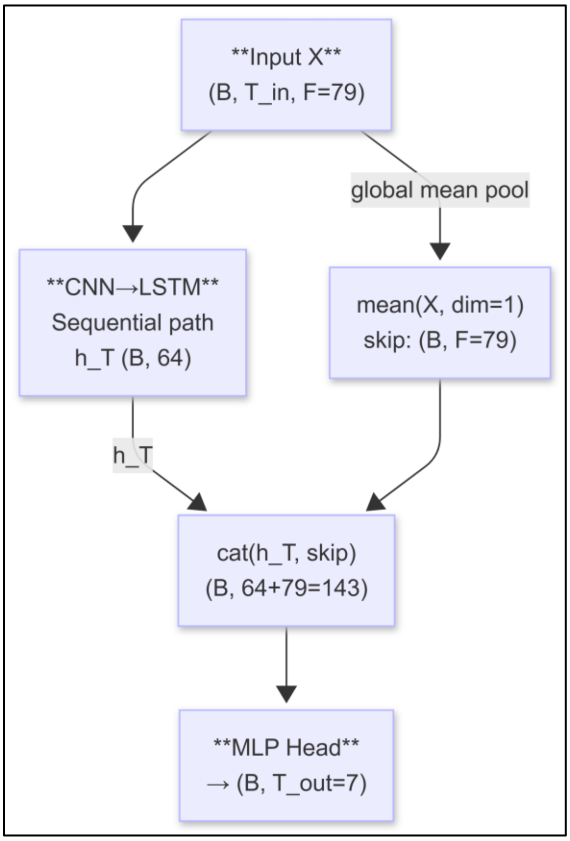
Sequential Mode: CNN extracts local temporal features, which the LSTM then processes for global sequential context. LSTM input is CNN output, not raw features.



Parallel Mode: CNN and LSTM process raw input independently; outputs concatenated.



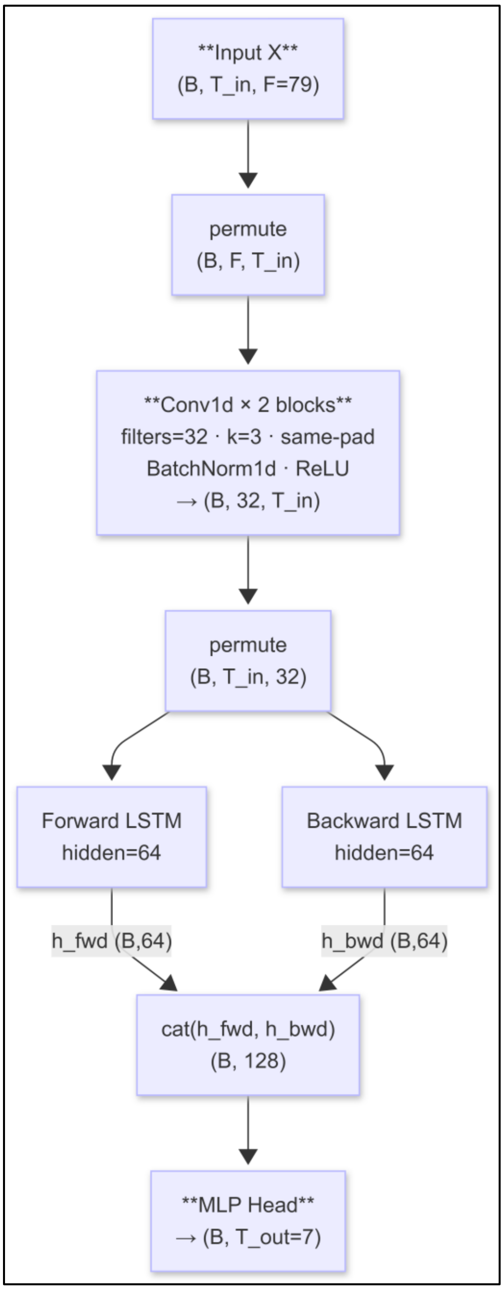
Augmented Mode: Sequential CNN-LSTM with raw-input skip connection to prevent information loss from CNN compression.



Hyperparameter changes for all three tuned variants:

- cnn_filters 32 → 64
- dropout 0.10 → 0.20
- Each mode is trained and evaluated independently
- Structure unchanged

CNN-BiLSTM

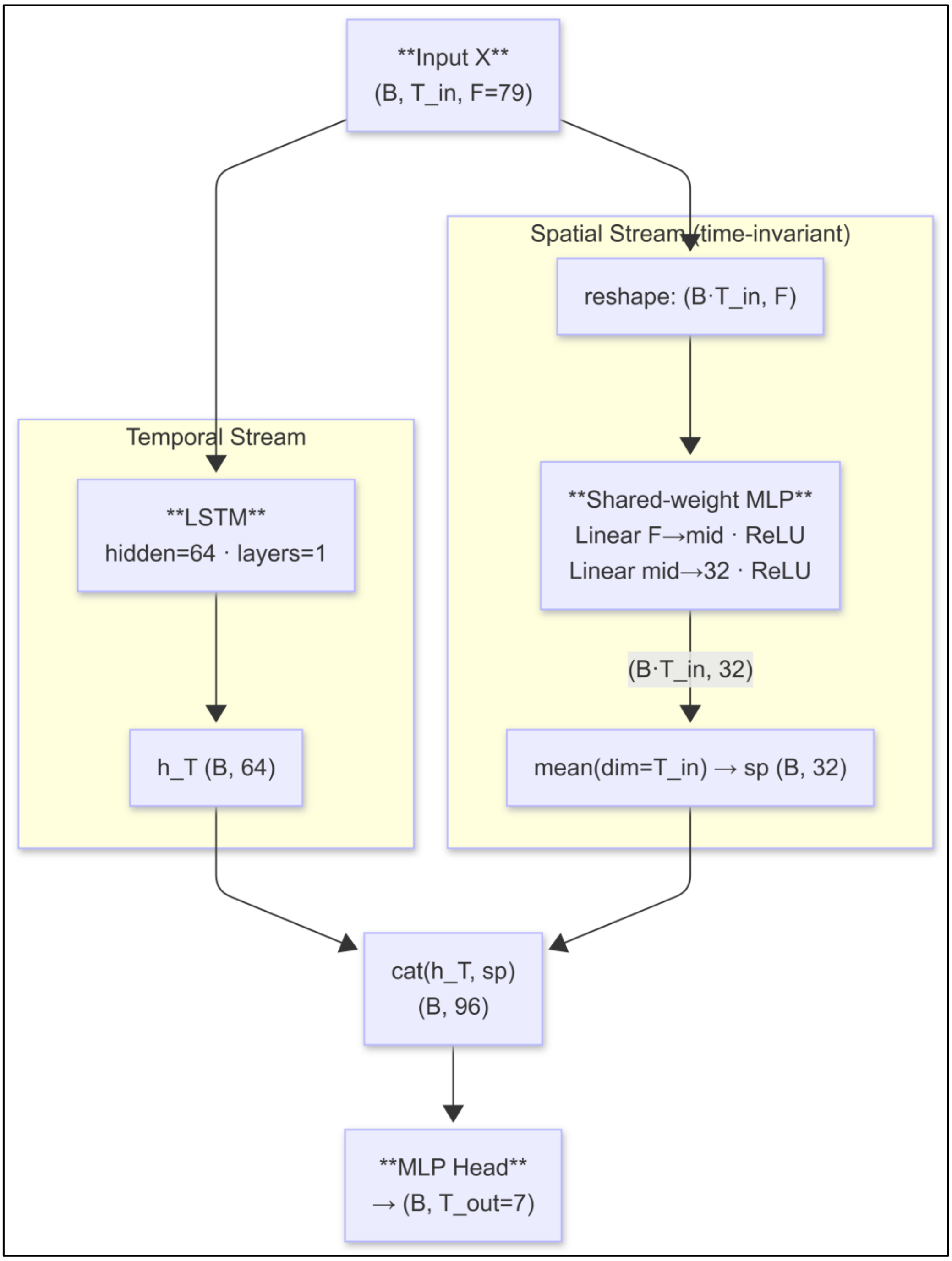


Hyperparameter changes for tuned variant:

- hidden $64 \rightarrow 128$
- cnn_filters $32 \rightarrow 64$
- cnn_layers $2 \rightarrow 1$
- dropout $0.10 \rightarrow 0.25$
- Structure unchanged

ST-LSTM – Spatio-Temporal LSTM

Two parallel streams are computed independently and concatenated before the MLP head. The spatial MLP is weight-shared across all T_{in} timesteps (time-invariant): it captures which features consistently co-activate in the window rather than when. Mean pooling over the time axis collapses temporal order to product a persistent cross-feature summary.



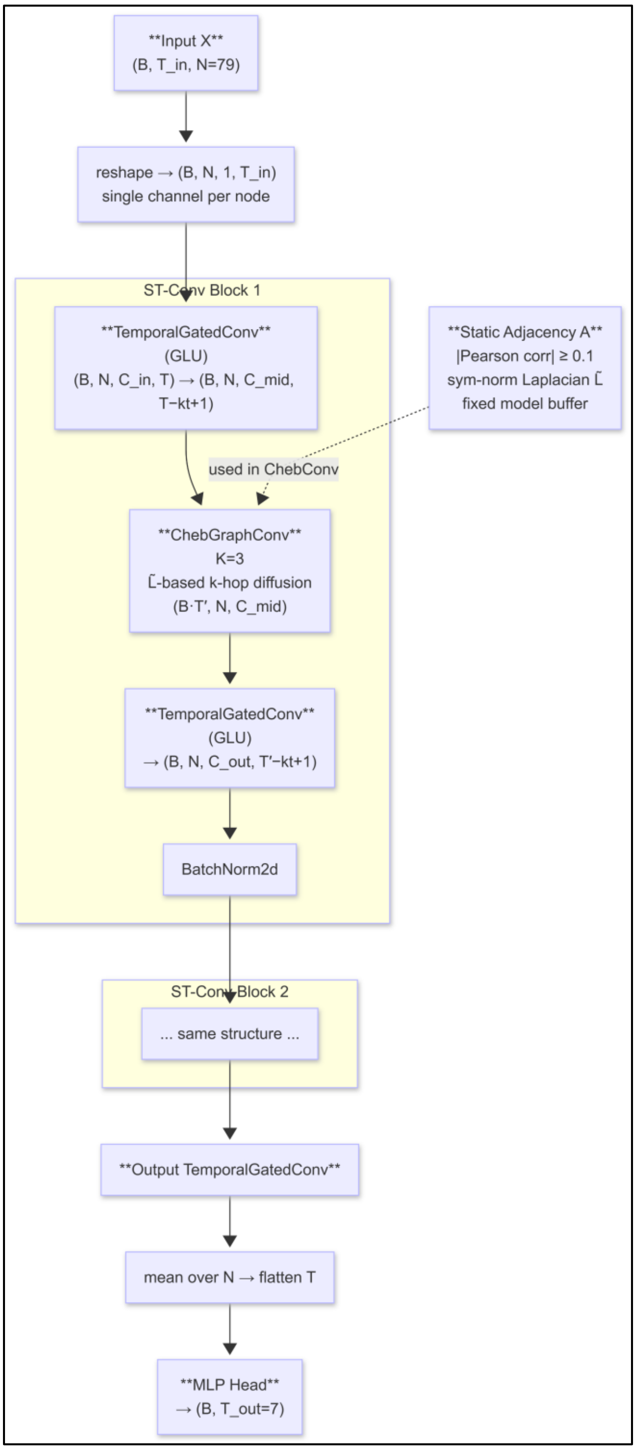
Hyperparameter changes for tuned variant:

- hidden 64 → 128
- spatial_hidden 32 → 128
- dropout 0.10 → 0.30
- Structure unchanged

APPENDIX C2 – Graph Models

STGCN – Spatio-Temporal Graph Convolutional Network

Input features are treated as N = 79 graph nodes each carrying a scalar temporal signal over T_{in} timesteps. The architecture alternates temporal gated convolutions (GLU) and Chebyshev graph convolutions in stacked ST-Conv blocks, with BatchNorm after each block.

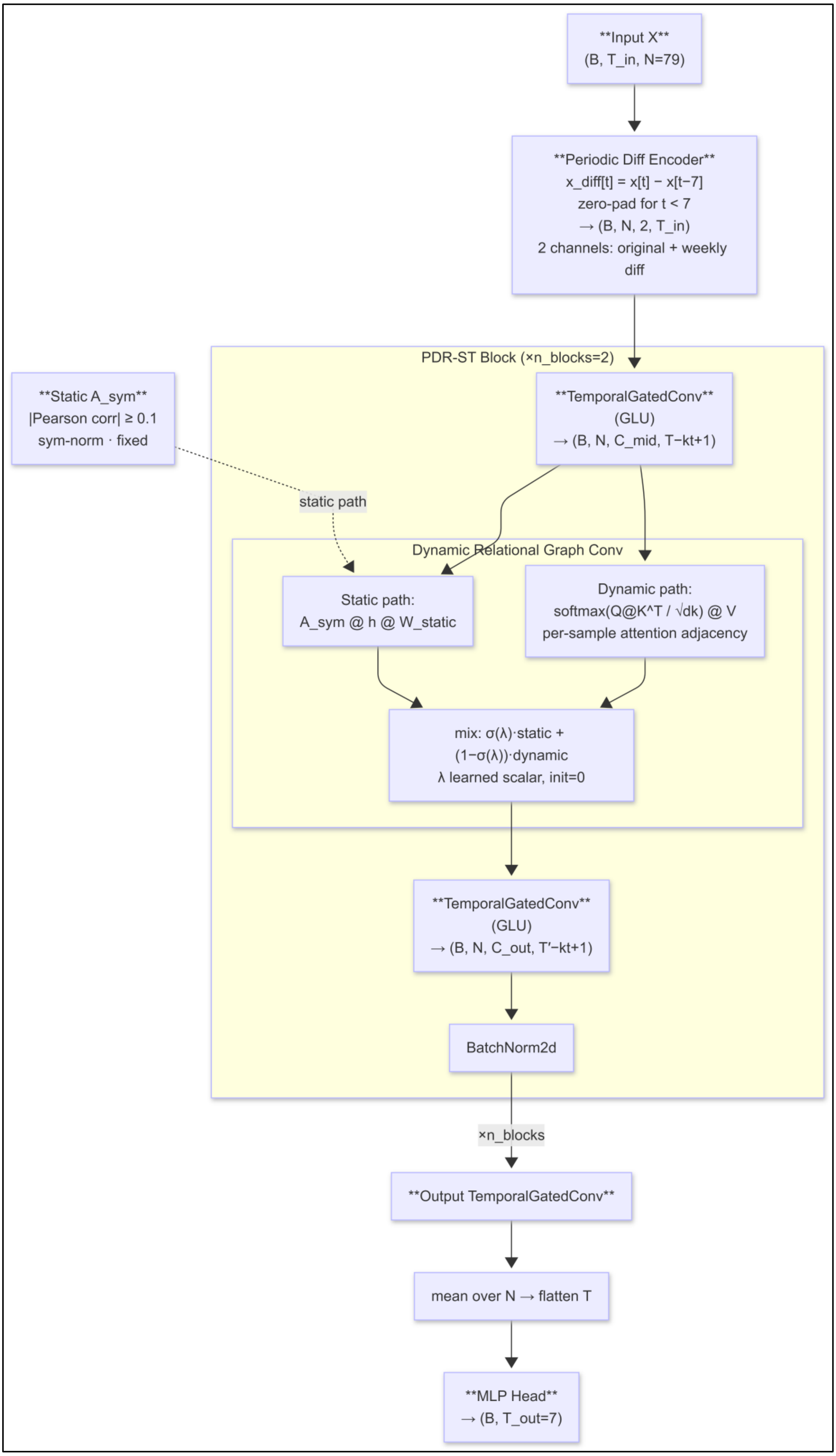


Hyperparameter changes for tuned variant:

- hidden $128 \rightarrow 256$
- dropout $0.20 \rightarrow 0.25$
- weight_decay $2e - 4 \rightarrow 3e - 4$
- Block depth and temporal kernel unchanged
- Structure unchanged

PDR-STGCN – Periodicity-Aware Dynamic Relational STGCN

A two-channel input (original signal + weekly lag-difference) feeds a modified STGCN backbone where each graph convolution replaces the fixed Chebyshev adjacency with a mixed static/dynamic adjacency controlled by a learned scalar λ.

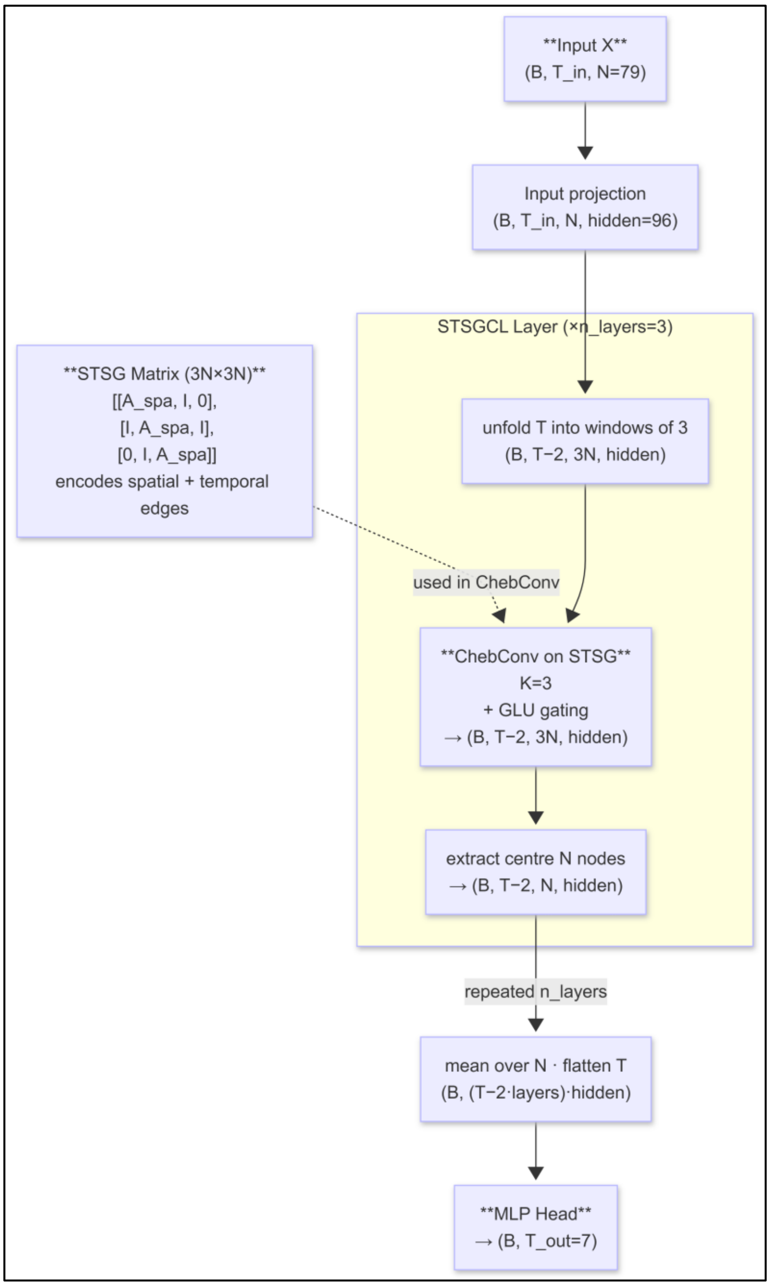

Input X
(B, T_in, N=79)
Periodic Diff Encoder
x_diff[t] = x[t] − x[t−7]
zero-pad for t < 7
→ (B, N, 2, T_in)
2 channels: original + weekly diff
PDR-ST Block (×n_blocks=2)
Static A_sym
|Pearson corr| ≥ 0.1
sym-norm · fixed
TemporalGatedConv
(GLU)
→ (B, N, C_mid, T−kt+1)
static path
Dynamic Relational Graph Conv
Static path:
A_sym @ h @ W_static
Dynamic path:
softmax(Q@K^T / √dk) @ V
per-sample attention adjacency
mix: σ(λ)·static +
(1−σ(λ))·dynamic
λ learned scalar, init=0
TemporalGatedConv
(GLU)
→ (B, N, C_out, T′−kt+1)
BatchNorm2d
×n_blocks
Output TemporalGatedConv
mean over N → flatten T
MLP Head
→ (B, T_out=7)

Hyperparameter changes for tuned variant:

- hidden $160 \rightarrow 256$
- n_blocks $2 \rightarrow 3$
- kt $3 \rightarrow 2$
- dk $48 \rightarrow 64$
- dropout $0.15 \rightarrow 0.25$
- Structure unchanged

<u>STSGCN – Spatial-Temporal Synchronous Graph Convolutional Network</u>

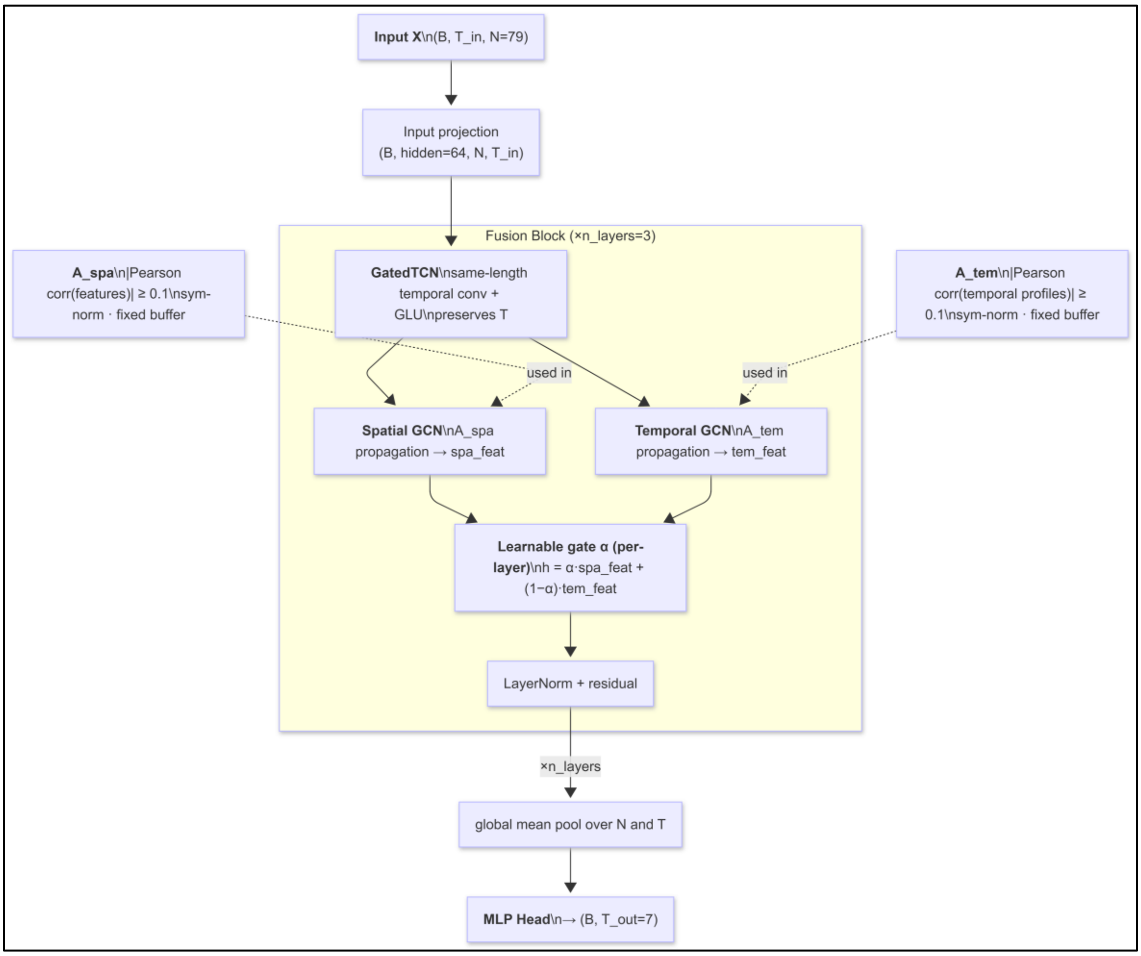
The $3N \times 3N$ synchronous graph (STSG) encodes both spatial adjacency (A_{spa} within each timestep) and temporal adjacency (identity block I linking adjacent timesteps). Each STSGCL layer applies Chebyshev convolution on the full STSG over a sliding 3-timestep window and then extracts the centre N nodes.



Hyperparameter changes for tuned variant:

- hidden 96 → 128
- dropout 0.10 → 0.25
- weight_decay $1e-4 \rightarrow 2e-4$
- lr $5e-4 \rightarrow 1e-3$
- Layer depth unchanged
- Structure unchanged

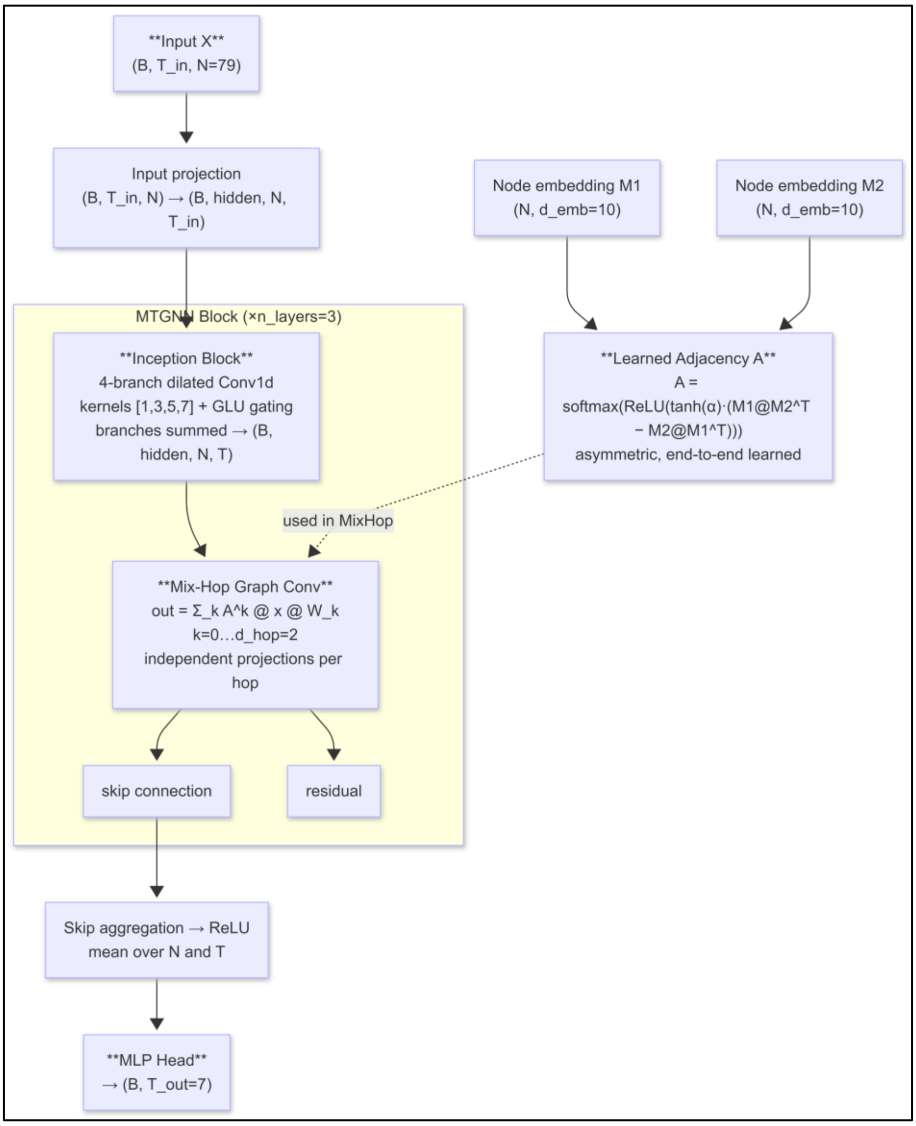
STFGNN – Spatio-Temporal Fusion Graph Neural Network



Hyperparameter changes for tuned variant:

- hidden 64 → 128
- dropout 0.20 → 0.35
- adj_threshold 0.10 → 0.15
- Layer depth is unchanged
- Structure unchanged

MTGNN – Multi-scale Temporal Graph Neural Network



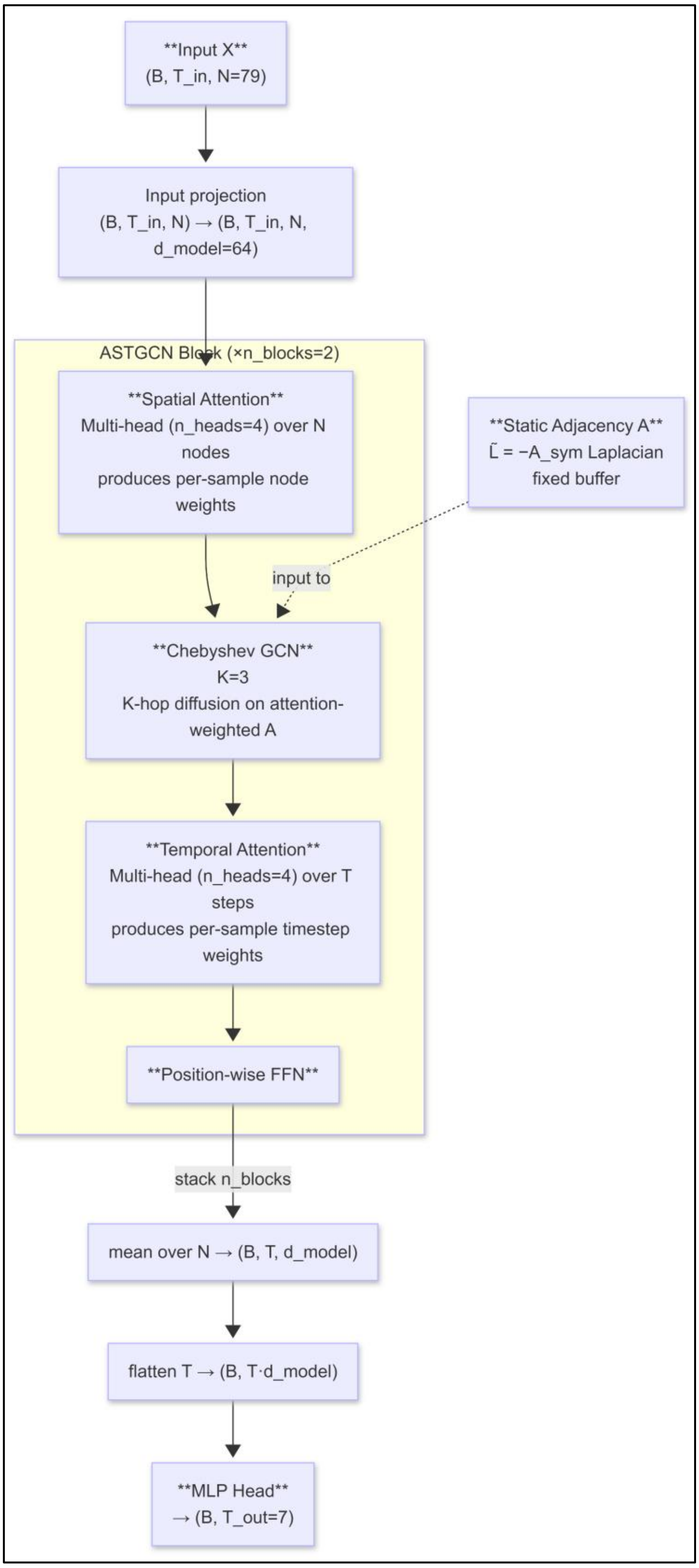
Hyperparameter changes for tuned variant:

- hidden 32 → 64
- d_emb 10 → 7
- dropout 0.10 → 0.30
- weight_decay $1e-4 \rightarrow 3e-4$
- Skip channels and layer depth unchanged
- Structure unchanged

APPENDIX C3 – Attention Models

ASTGCN – Attention-based STGCN

Use AMP (fp16 + GradScaler) for memory efficiency.

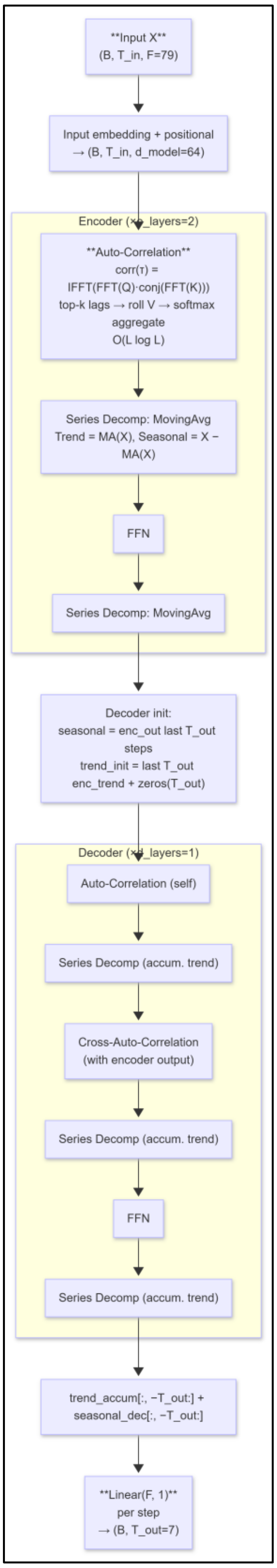


Hyperparameter changes for tuned variant:

- d_model $64 \rightarrow 128$
- n_heads $4 \rightarrow 8$
- n_blocks $2 \rightarrow 3$
- dropout $0.15 \rightarrow 0.20$
- weight_decay $2e-4 \rightarrow 1e-4$
- Structure unchanged

Autoformer

FFT operations are explicitly cast to float32 inside autocast for numerical stability under AMP.

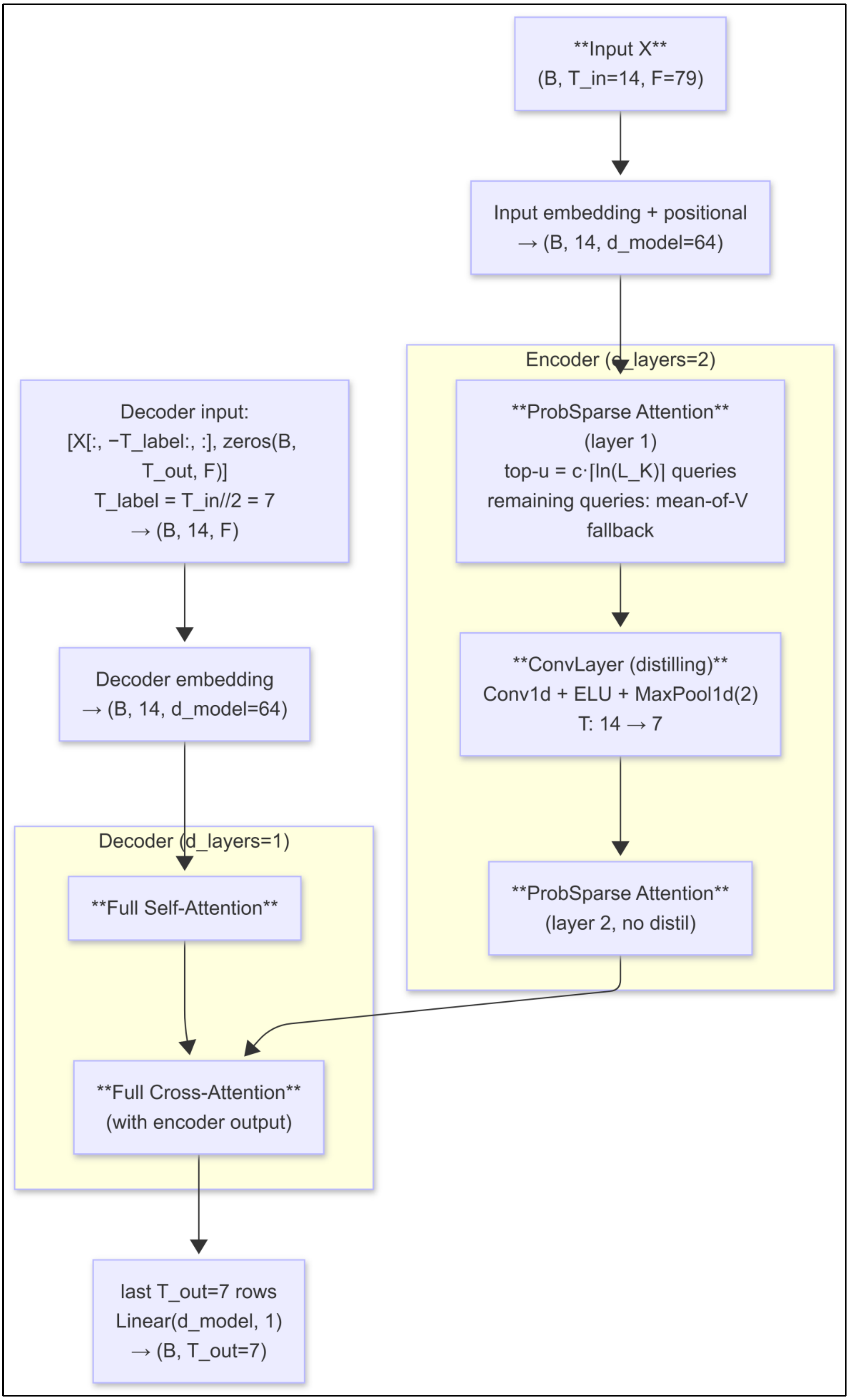


Hyperparameter changes for tuned variant:

- d_model $64 \rightarrow 128$
- n_heads $4 \rightarrow 8$
- e_layers $2 \rightarrow 3$
- d_ff $128 \rightarrow 256$
- dropout $0.15 \rightarrow 0.25$
- Structure unchanged

Informer

Uses AMP (fp16 + GradScaler). At T_{in} = 14, the single distilling step halves the encoder sequence to length 7, and ProbSparse selects approximately 13 of 14 queries (near-full attention).



Input X
(B, T_in=14, F=79)
Input embedding + positional
→ (B, 14, d_model=64)
Encoder (e_layers=2)
ProbSparse Attention
(layer 1)
top-u = c·⌈ln(L_K)⌉ queries
remaining queries: mean-of-V
fallback
ConvLayer (distilling)
Conv1d + ELU + MaxPool1d(2)
T: 14 → 7
ProbSparse Attention
(layer 2, no distil)
Decoder input:
[X[:, −T_label:, :], zeros(B, T_out, F)]
T_label = T_in//2 = 7
→ (B, 14, F)
Decoder embedding
→ (B, 14, d_model=64)
Decoder (d_layers=1)
Full Self-Attention
Full Cross-Attention
(with encoder output)
last T_out=7 rows
Linear(d_model, 1)
→ (B, T_out=7)

Hyperparameter changes for tuned variant:

- d_model 64 → 256
- n_heads 4 → 16
- e_layers 2 → 3
- d_ff 128 → 512
- Dropout unchanged
- Structure unchanged

APPENDIX C4 – HMT-TSF Training Configurations

- Epochs: 150
- Batch size: 32
- Optimiser: AdamW (lr = 1e-3, weight_decay = 1e-3)
- LR schedule: Linear warmup (8 ep) → ReduceLROnPlateau (factor = 0.5, patience = 8)
- Gradient clip: max_norm = 1.0
- Early stopping: patience = 20 (raw val loss)
- Mixed precision: AMP fp16 on CUDA (GradScaler)
- Loss: WeightedHuber (loss_decay = 0.9) + TemporalSmoothness (smooth_weight = 0.01)
- Input noise: Gaussian noise std = 0.02 added to training inputs (data augmentation)
- d_model: 64
- n_tcn_blocks: 3
- tcn_kernel: 3
- graph_hidden: 64
- n_regimes: 3

- n_attn_heads: 4
- adj_threshold: 0.1
- drop_path: 0.2
- dropout: 0.1
- Optional residual boosting: CatBoost or sklearn MLP on train-set residuals; applied at 0.5x weight only if it improves Combined%
- Walk-forward evaluation: test set split into 3 equal blocks; per-block Combined% and R^2 reported.